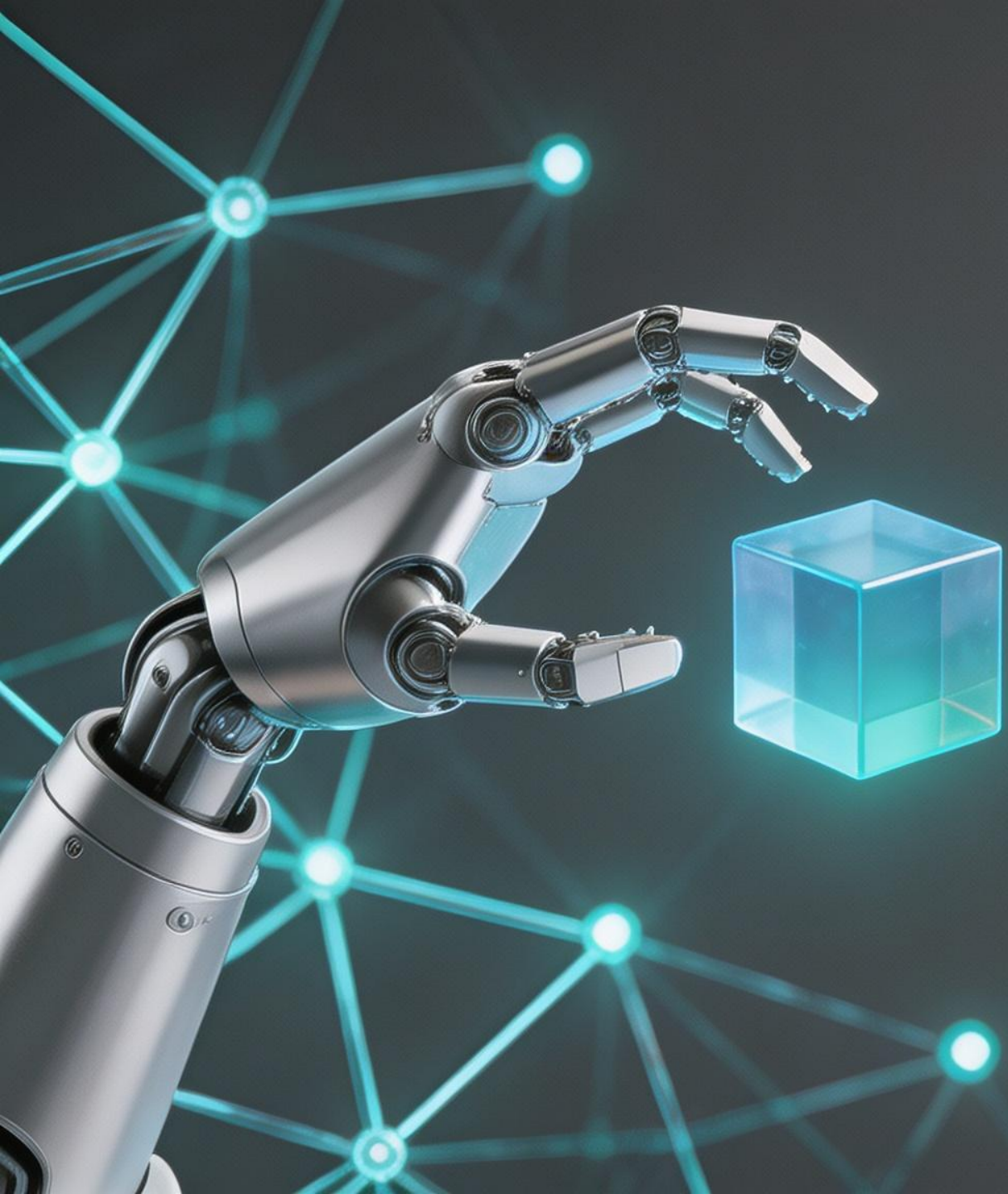




Fundamentals of Robot Learning

A Comprehensive Guide

Presenter: Mengyuan Wang

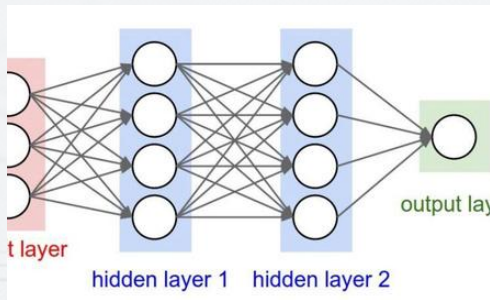
Date: 2026/08/15

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1. Robot Learning vs. Ordinary Machine Learning

Ordinary Machine Learning (OML)



Focuses on **static pattern recognition** from pre-collected, curated datasets. It operates within a closed, digital realm

with no real-world physical



Static, Passive Data: Uses independent, identically distributed (i.i.d.) datasets with no temporal dependency.



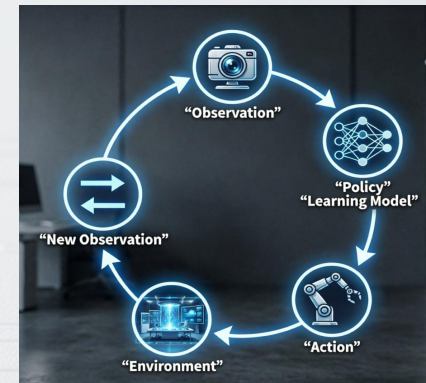
Open-Loop Process: Predictions do not influence the

input data; a one-way information flow.



Abundant Data: Relies on large, readily available digital datasets with low cost of collection.

Robot Learning (RL)



Physically embodied and environmentally situated:

Emphasizes learning by doing through direct interaction with a

dynamic physical environment

Interactive, Sequential Data: Data is generated by the robot's actions, creating a causal loop between decision



and outcome.



Closed-Loop Dynamics: Actions alter the environment, which in turn provides new observations for the next step.



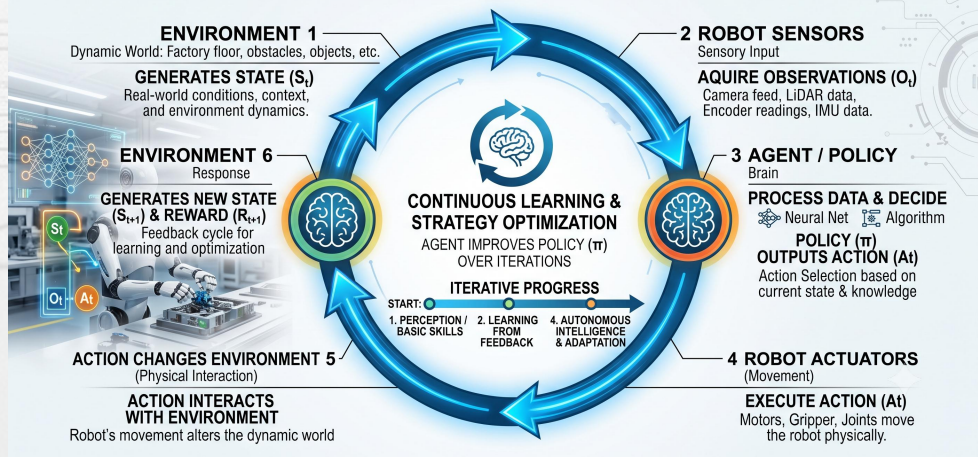
Safety-Critical & Efficient: High stakes for failure.

2. Closed-Loop Sequence: From Observation to Action



CLOSED-LOOP ITERATIVE LEARNING PROCESS FOR ROBOTIC INTELLIGENCE EVOLUTION

From Perception to Autonomous Intelligence through Continuous Practice and Optimization



Closed-Loop Iterative Learning Framework for Robotic

Intelligence Evolution

State (S_t)— The complete, objective configuration of the environment, which is often hidden and unobservable directly.

Action(a_t)— Action represents the specific control commands executed by a robot, such as joint targets or velocities.



$$a_t = \pi_{\theta}(o_t)$$

$$a_t = \pi_{\theta}(o_{t-H:t})$$

Observation (O_t)— Partial, sensor-derived data (e.g., camera feeds) that is often noisy and incomplete.

Policy (π_{θ})— The robot's "brain": a mapping function that translates observations into optimal actions.

$$a_t = \pi_{\theta}(o_{1:t})$$

2.1 The Critical Difference: State vs. Observation



Observation $\neq$ State



State (s_t) — The Ground Truth

The complete, objective reality of the environment — every physical detail, force, and entity in perfect, unfiltered clarity. This is the **omniscient perspective** that no real-world system can ever measure directly.

Example: In self-driving, this means knowing the exact speed, acceleration, and intent of every car, pedestrian, and cyclist within 1 km — including those hidden around corners or obscured by other vehicles.

💡 **Inaccessible in reality, existing only in theory or simulation.**



Observation (o_t) — The Robot's View

The imperfect, sensor-derived snapshot of the world. Observations are inherently noisy, limited by hardware resolution, and spatially incomplete — **missing information hidden from the robot's field of view.**

Example: Blurry camera images with lens flare, a sparse LiDAR point cloud missing small objects, or delayed GPS data. A pedestrian blocked by a truck would not appear in the robot's observations at all.

⚠️ **The sole basis for decision-making, and the primary source of uncertainty.**



The Core Challenge: Robots can never access the "true state." They must learn to make optimal decisions based solely on flawed, incomplete observations. Bridging this gap is the fundamental problem of perception and control in robotics.

2.2 Observation



Hardware Foundation: The physical basis for perception and action, and the first line of defense against observation challenges.

01 Sensing Modalities

- **Visual Perception:** RGB provides semantics and texture, while RGB-D and point clouds construct 3D geometry.
- **Proprioception:** Joint angles and torque feedback form the foundation of motion control.
- **Force/Tactile Sensing:** critical for delicate tasks such as assembly and compliant grasping.

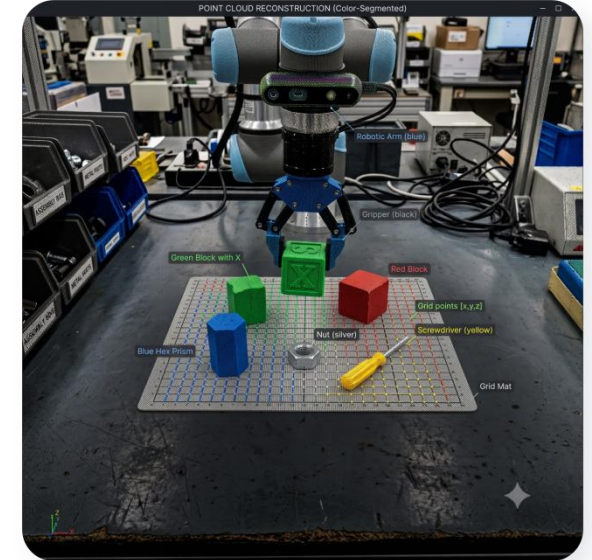
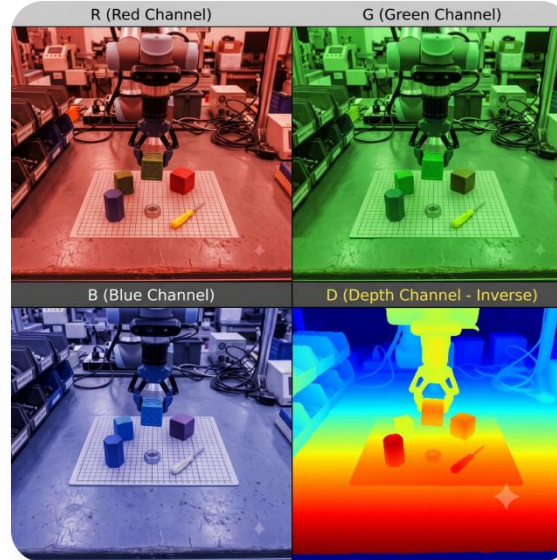
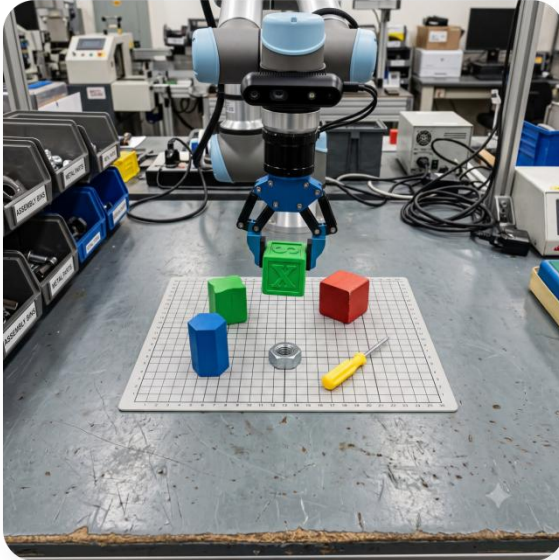
02 Key Challenges

- Partially Observable
- Occlusion
- Noise
- Latency
- Multi-view Fusion
- Coordinate Transformation
- History Dependence

03 POMDP

Real-world robotic tasks are generally better modeled as POMDPs rather than fully observable MDPs.

2.2.1 Visual Perception: RGB, RGB-D, and Point Clouds



01 RGB

Advantages: Low cost, rich in color and texture details, and highly compatible with mature pretrained vision models for semantic understanding.

Limitations: Merely a 2D projection lacking 3D depth information, and highly sensitive to environmental factors such as lighting changes and object occlusion.

02 RGB-D

Advantages: Simultaneously captures appearance and geometric depth, providing critical 3D spatial references for autonomous navigation, dynamic obstacle avoidance, and object grasping.

Limitations: Inherent noise in depth data, limited measurement accuracy for transparent or highly reflective materials, and a relatively restricted effective detection range.

03 Point Clouds

Advantages: Directly represents the geometric structure of 3D space with the richest detail, making it an ideal input for high-precision mapping and complex manipulator operations.

Limitations: Massive unstructured data leads to extremely high computational and storage costs, along with high algorithmic complexity for data denoising, registration, and feature extraction.

2.2.2 Proprioception

The perception of a robot's own state is the core foundation for precise motion control, autonomous localization, and intelligent decision-making.



01 Joint Angles

Provides real-time feedback on the current absolute or relative position of each servo joint.



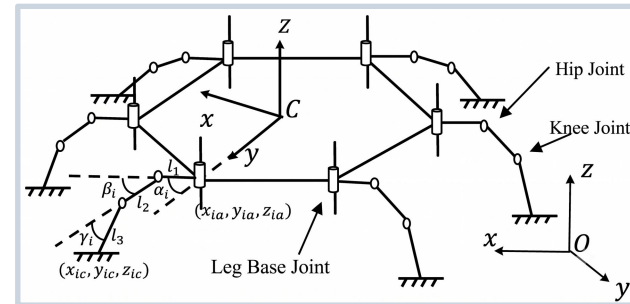
02 Joint Velocity

Monitors changes in the angular velocity of joint rotation, enabling smooth trajectory planning.



03 Joint Torque

Senses the force and torque output at the joints to ensure operational safety.



Multi-Joint Coordinated Perception

Through high-precision sensing of each joint, the robot can construct a digital "skeleton map" of its own body.



04 End-Effector Pose

Determines the precise coordinates and orientation of the manipulator's end-effector in 3D space.



05 Gripper State Perception

Provides real-time feedback on the opening/closing degree and gripping force.



Core Value: The Foundation of Autonomous Learning for Robots — Without proprioceptive state perception, a robot would lose control over its own limbs, rendering it incapable of any autonomous decision-making or delicate manipulation.

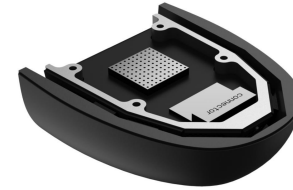
2.2.3 Force/Tactile Sensing

In delicate tasks such as assembly and polishing, visual information alone is often insufficient to determine contact status and force conditions.



01 Force/Torque

Core Function: Real-time measurement of interaction forces and torques between the robot and the environment, providing precise force feedback to prevent excessive or insufficient force.



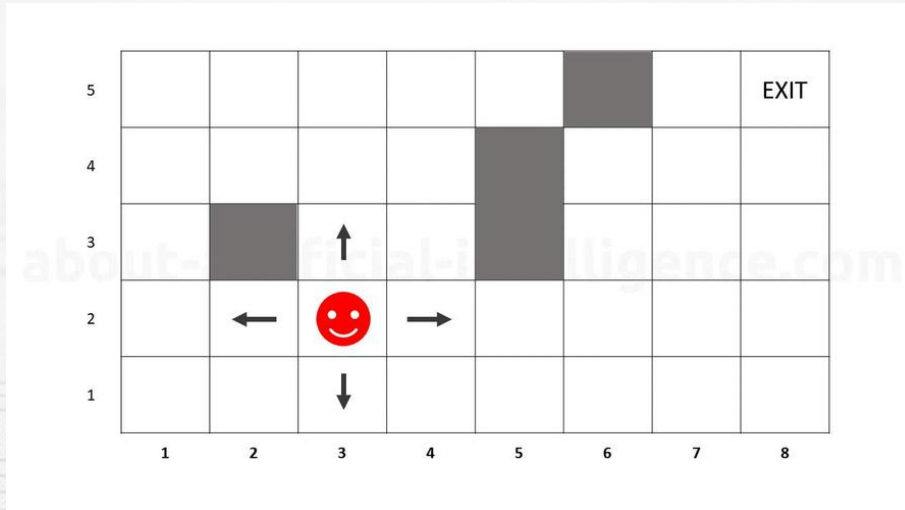
02 Tactile

Core Function: Perception of pressure distribution, micro-deformations, and object slippage trends at the contact area, and even recognition of surface textures and materials.



Core Insight: Combining vision and tactile sensing allows the robot not only to "see" but also to "feel and precisely control," marking a crucial step toward intelligent compliant manipulation.

3. Core Concepts: Trajectory, Reward, Demonstration



This grid world example illustrates how an agent navigates an environment. Every step involves a **Policy** decision, contributing to a **Trajectory**, guided by the **Reward** signal towards the goal, or learned from expert **Demonstrations**.

Trajectory/episode/rollout (τ)

The complete story of an episode: $\tau = (o_1, a_1, o_2, a_2, \dots, o_T, a_T)$. Each path has an associated return $G(\tau)$, which is the cumulative sum of rewards obtained along the path.

Reward ($r_t = r(o_t, a_t)$)

A scalar value evaluating the desirability of an action. Designing a good reward function is critical but often challenging, especially with sparse or deceptive rewards.

Demonstration ($\mathcal{D} = \{\tau_i\}_{i=1}^N$)

A dataset collected from an expert performing the task, allowing the agent to learn by mimicking successful behavior.

4.1 What is Degrees of Freedom (DoF)?

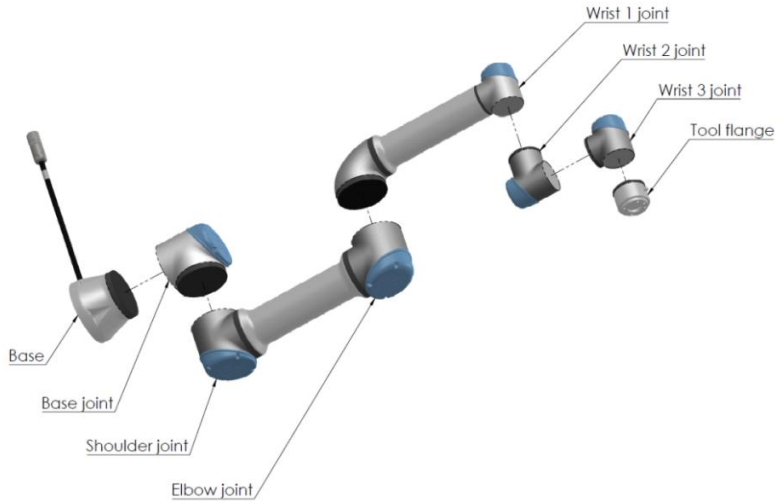


Figure: UR5e Robot Arm – Joints, Base, and Tool Flange

Definition

The count of independent motion parameters defining a robot's state.

Intuitive Insight: Dimensions from 1 to 6

A simple hinge joint has only 1 DoF; whereas a free rigid body possesses the full 6 DoF in 3D space, comprising 3 translational degrees of freedom along the X, Y, and Z axes, and 3 rotational degrees of freedom around the three axes (roll, pitch, and yaw).

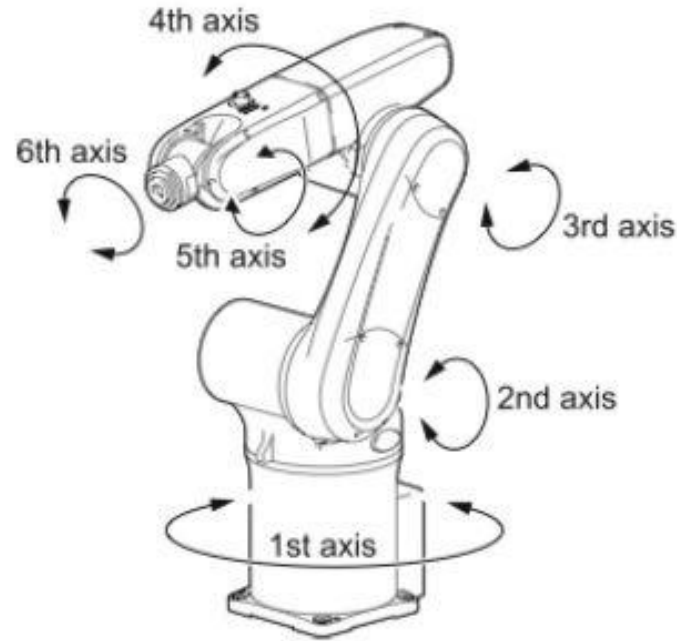


Figure: A standard 6-axis articulated robotic arm, highlighting its rotational degrees of freedom which enable precise manipulation in 3D space.

Applications and Classification in Robotics

Joint Degrees of Freedom: In a standard serial manipulator, each independently driven single-DoF joint contributes one actuated degree of freedom to the system.

End-Effector Degrees of Freedom: Refers to the pose (6D Pose) that the end-effector can achieve in space, which directly determines the robot's workspace and dexterity.

4.2 Redundant DoF, Underactuation, and High-Dimensional Control



Figure: A typical collaborative operation scenario of multi-DoF industrial manipulators.

01 Redundant DoF

- Redundancy: **Joint DoF > Task DoF** (e.g., 7-DoF arm for 6-DoF end-effector).
- Advantages: Enhanced obstacle avoidance & reachability.
- Limitations: Complex inverse kinematics & control design.

02 Underactuation

- Underactuation: **Actuators < DoF** (Common in humanoids & UAVs).
- Core: Exploit dynamic coupling for indirect control of unactuated DoFs.

03 Challenges and Opportunities in High-Dimensional Control

- Challenge: **DoF ↑ → State Space Expands Exponentially** (High compute & data demand).
- Opportunity: Richer locomotion & better adaptability.

4.3 Action Space



01 Discrete Action Space

High-Level Decision (Multiple-Choice):

- ✓ Action: Select from discrete options (e.g., pick, place).
- ✓ Algorithm: Classification.



02 Continuous Action Space

Low-Level Control (Fill-in-the-Blanks):

- ✓ Action: Output continuous values ($a_t \in \mathbb{R}^{d_a}$) for joints/torque.
- ✓ Algorithm: Regression.



Engineering Insight:

Core: Continuous actions dominate for high-precision & high-dynamic control.

Discrete: High-level "trigger signals" only.

Continuous: Low-level execution relies on continuous numerical commands.

4.4 Joint Space Actions vs. Cartesian Space Actions

Dimension	Cartesian Space/Task Space	Joint Space
Focus	End-Effector / Task Target	Robot Arm Joints / Motors
Representation	Position (X, Y, Z) + Orientation (Roll, Pitch, Yaw) $a_t = [\Delta x, \Delta y, \Delta z, \Delta r_x, \Delta r_y, \Delta r_z, g]$	7DOF: $a_t = [q_1, q_2, \dots, q_7, g]$
Dimension Fixedness	Fixed at 6D (governed by the physical laws of 3D space)	Depends on the arm's structure (equal to the number of joints)
Intuitiveness	Highly intuitive for humans (e.g., "Move the gripper forward by 10 cm")	Extremely non-intuitive for humans (e.g., "Rotate Axis 1 by 10° , Axis 2 by -5° ")
Control Complexity	Requires complex mathematical transformations and suffers from singularity issues	Most direct for motor actuation

4.5 Control Modes: Position, Velocity, Torque, and Impedance

The same learned policy behaves differently under different robot control interfaces.

Higher level,
more stable



Position Control

Command: Target Position

Advantages: High rigidity & precision.

Limitations: Unsafe for hard contact (high impact).



Velocity Control

Command: Target Velocity

Advantages: Smooth motion, fast response.

Limitations: Weak position holding, drift issues.



Torque Control

Command: Target Torque

Advantages: Highest bandwidth, direct force interaction.

Limitations: Unstable open-loop, hard to tune.

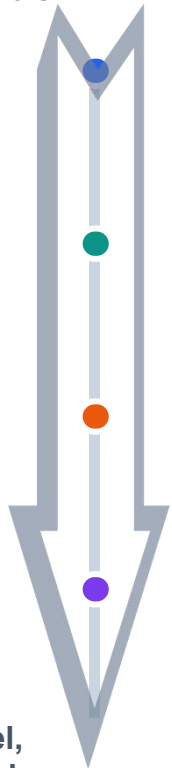


Impedance Control

Command: Stiffness & Damping

Advantages: Safe & compliant interaction (spring-damper).

Limitations: Complex tuning, lower precision than position control.



Lower level,
more flexible



Takeaway: Position and velocity are common in imitation learning; torque is lower level; impedance bridges learned policies and real contact.

5. Main Learning Paradigms: RL vs. IL



Figure: Expert demonstrations via VR/AR teleoperation enable efficient skill transfer to the robotic arm.

01. Reinforcement Learning (RL) – Trial-and-Error

- ◆ Core: Optimize behavior via environment rewards $\max_{\theta} \mathbb{E}_{\pi_{\theta}} [\sum_{t=1}^T \gamma^{t-1} r_t]$.
- ◆ Advantages: Discovers novel, super-human strategies.
- ◆ Limitations: Sample-inefficient, credit assignment & exploration-exploitation dilemmas.

02. Imitation Learning (IL) – Learning by

Demonstration

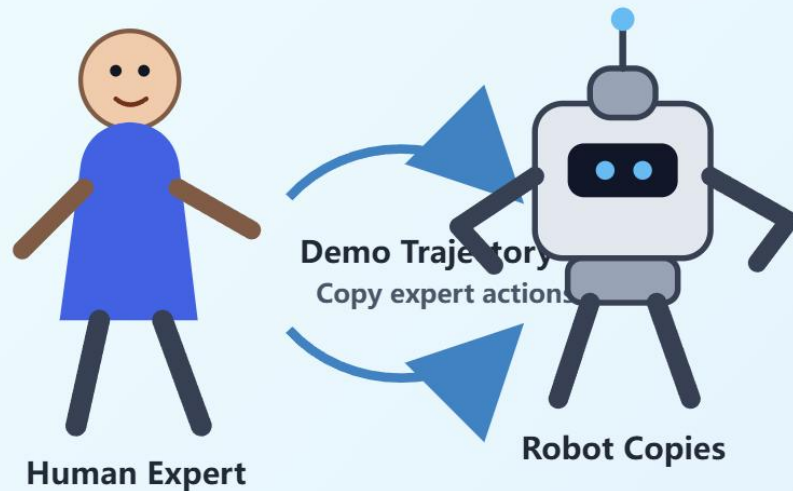
- ◆ Core: Mimic expert behavior.
- ◆ Advantages: Sample-efficient, no complex reward design.
- ◆ Limitations: Limited by expert, error compounding & covariate shift.

5. Main Learning Paradigms: RL vs. IL

Imitation Learning

Learn by watching expert demonstrations

Expert Data

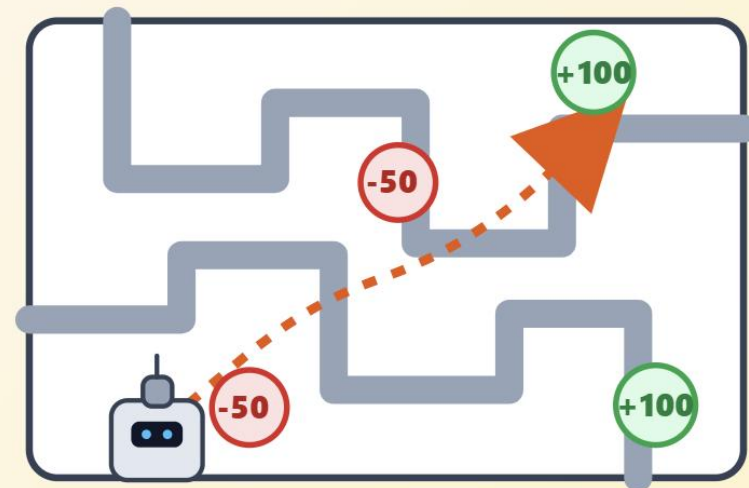


Expert actions → Supervised learning → Policy

Reinforcement Learning

Learn by trial-and-error to maximize return

Environment Feedback



Action → Feedback → Policy update

6. Behavior Cloning (BC)

Behavior Cloning (BC):

- Paradigm: Supervised Learning
- Mechanism: Direct observation-to-Action Mapping
- Key Feature: No Reward Signals Required

Core Mechanism: A 3-Step Process

 **01. Data Capture**

$$\mathcal{D} = \{(o_t, a_t)\}_{t=1}^N$$

 **02. Supervised Training**

$$a_t \approx \pi_{\theta}(o_t)$$

Continuous Action Space:

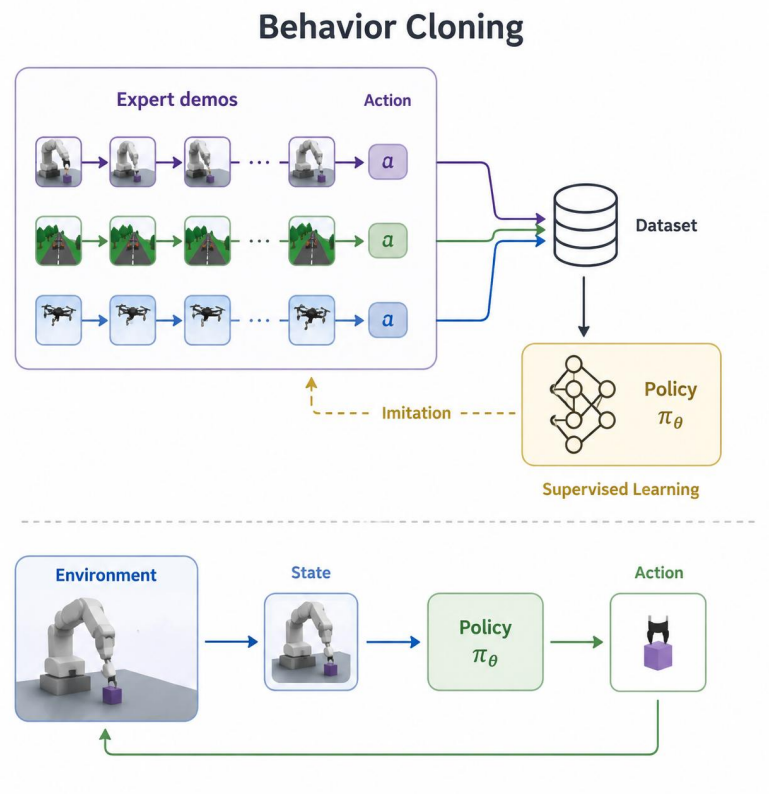
$$\min_{\theta} \sum_t \|\pi_{\theta}(o_t) - a_t\|^2$$

Discrete Action Space:

$$\min_{\theta} - \sum_t \log \pi_{\theta}(a_t | o_t)$$

 **03. Policy Execution**

The trained model directly outputs actions $\pi_{\theta}(o_t)$ for the agent.



Catastrophic Compound Error

Small deviations accumulate, causing the agent to drift into unobserved states where the policy fails entirely.

Extremely Data-Hungry

Requires massive, diverse datasets to cover all edge cases, making it impractical for complex real-world tasks.

Performance Ceiling

The model can never outperform the expert data it was trained on, limiting its potential for autonomous improvement.

6.1 Core Issues in BC: Distribution Shift

Core Issue in Behavior Cloning: Distribution Shift

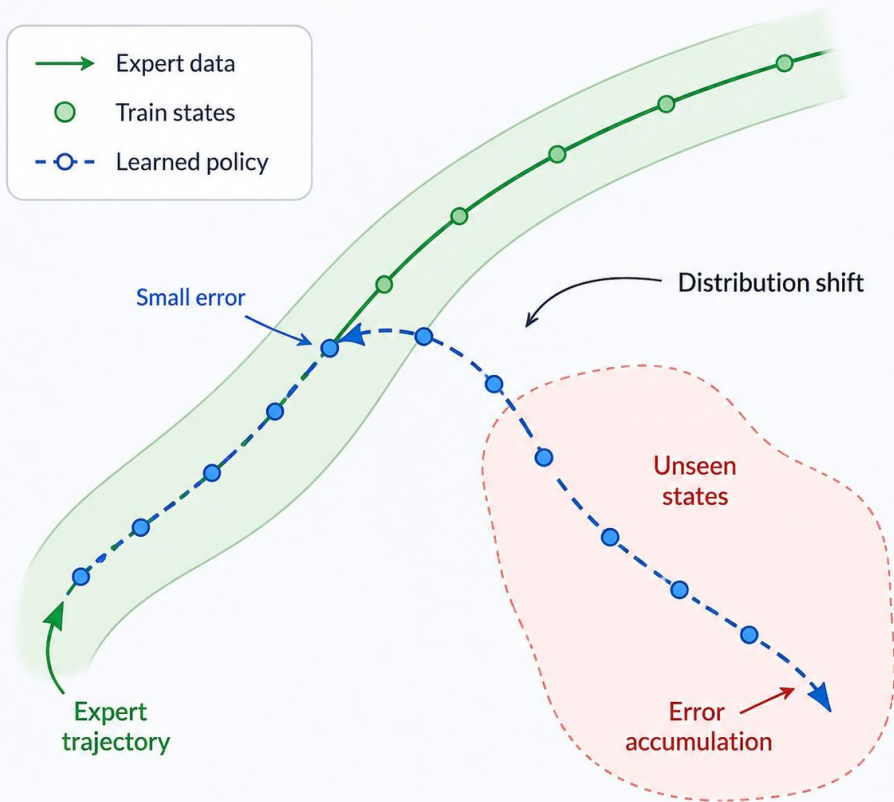


Figure: Distribution Shift

01. Root Cause: Train-Test Distribution Gap

- ✓ Training: Learns from expert's perfect states.
- ✓ Deployment: Faces its own real states.

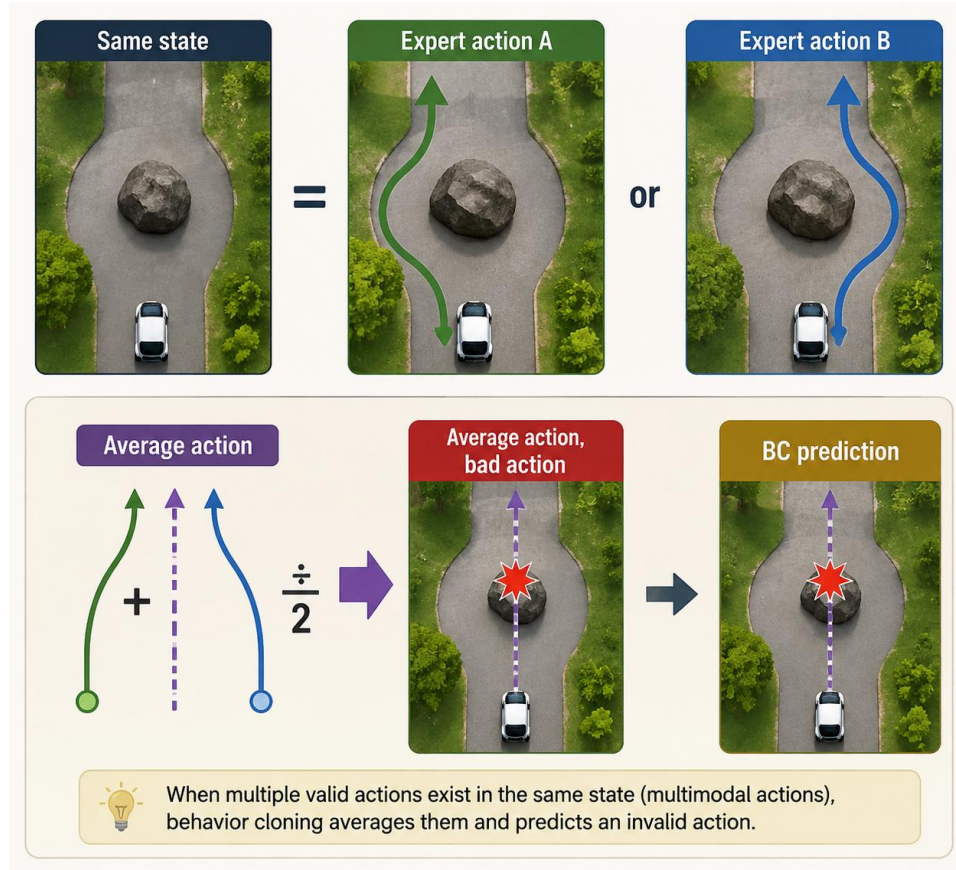
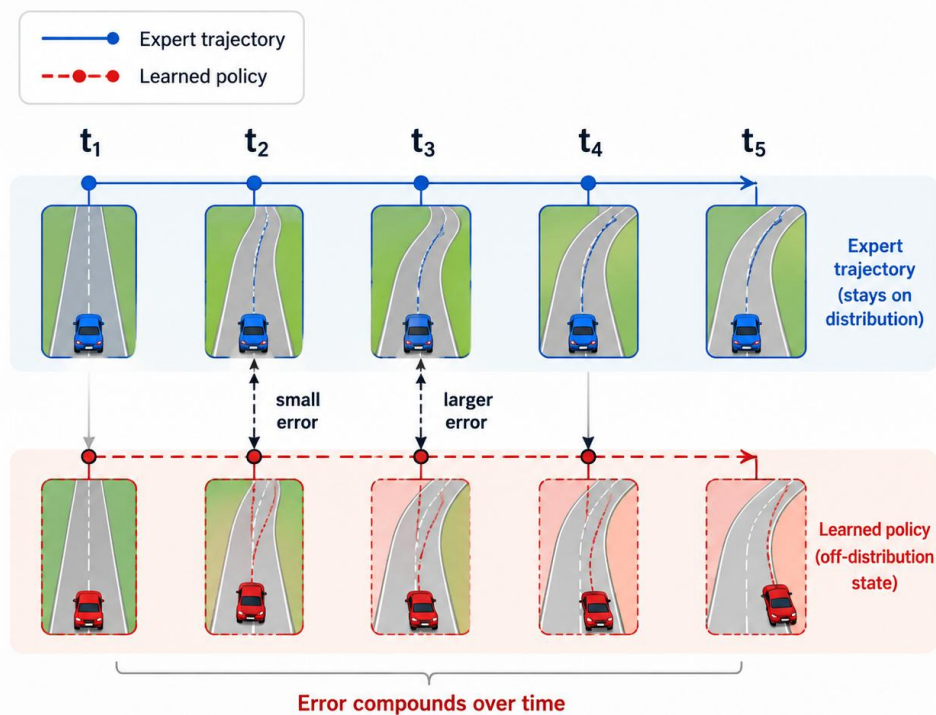
02. Cascading Shift Mechanism

- ① Tiny Gap: Slight policy difference creates initial deviation.
- ② Unknown States: Robot enters states unseen in training.
- ③ Error Amplification: Lack of data support causes exponential error growth.
- ④ Completely diverges from the reference trajectory.

03. Consequence: Rapid Task Failure

- ✓ Strategy quickly enters Out-of-Distribution (OOD) zones.
- ✓ No expert data to correct errors → Task fails rapidly.

6.2 Core Issues in BC: Compounding Error and Multimodal Actions



01 Compounding Error

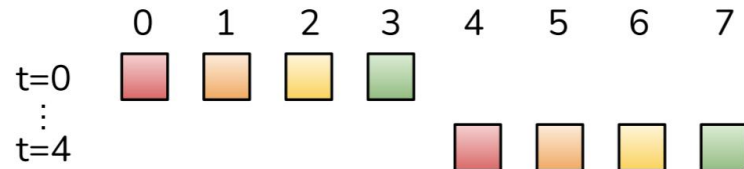
- **Mechanism:** Tiny prediction errors accumulate over long sequences.
- **Consequence:** Like a snowball, small deviations amplify exponentially, causing the system to completely diverge from the expected trajectory.

02 Multimodal Actions

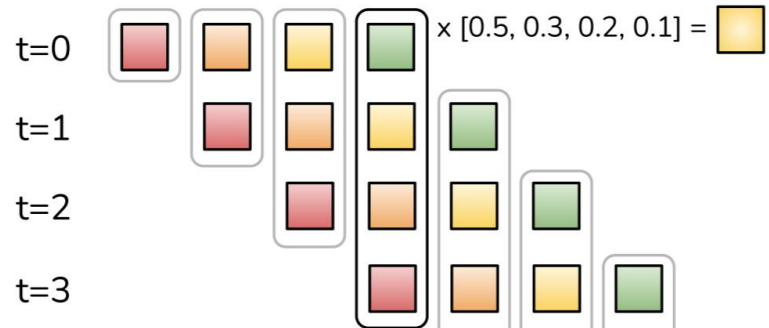
- **Mechanism:** Multiple valid paths exist for the same observation (e.g., going left or right around an obstacle).
- **Consequence:** Standard MSE regression loss fails to model this multimodal distribution, leading to "averaged-out" and often invalid actions.

7. Action Chunking

Action Chunking



Action Chunking + Temporal Ensemble



01 Issues with Single-Step Action Prediction

- ✓ **Frame-by-Frame:** Independent predictions can cause action jitter.
- ✓ **Error Accumulation:** High-frequency control amplifies errors over time.
- ✓ **Short-Sighted:** Lacks short-term intent (only predicts the next step).
- ✓ **Coordination Issues:** Unstable for dual-arm & high-DoF tasks.

02 Core Idea: Sequence Generation

$$\pi_{\theta}(o_t) \rightarrow a_t, a_{t+1}, \dots, a_{t+k-1}$$

Or:

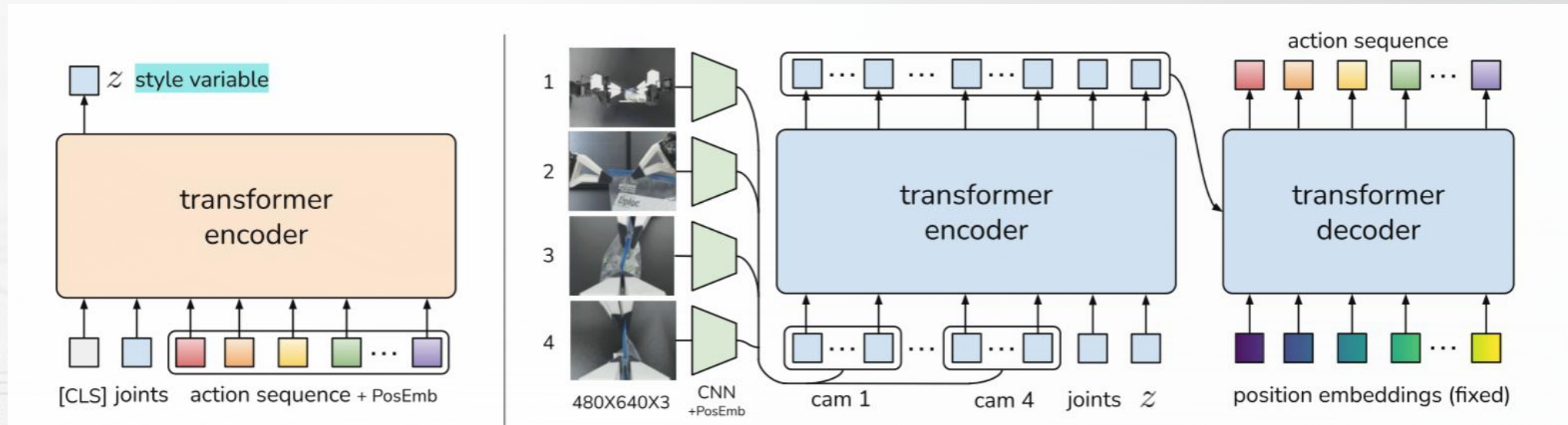
$$\pi_{\theta}(o_t) \rightarrow A_t$$

$$A_t = [a_t, a_{t+1}, \dots, a_{t+k-1}]$$

03 Action Horizon vs. Execution Horizon

- **Action Horizon:** Number of future steps predicted by the model.
- **Execution Horizon:** Number of steps actually executed.

8. ACT: Action Chunking with Transformers



Transformer Architecture

Processes multi-view images and proprioceptive states to efficiently model complex temporal dependencies.



Action Chunking

Predicts multiple consecutive future actions in a single pass, ensuring smooth and coherent manipulation.



CVAE Diversity Modeling

Introduces Conditional Variational Autoencoders to generate diverse action distributions.



Temporal Ensembling

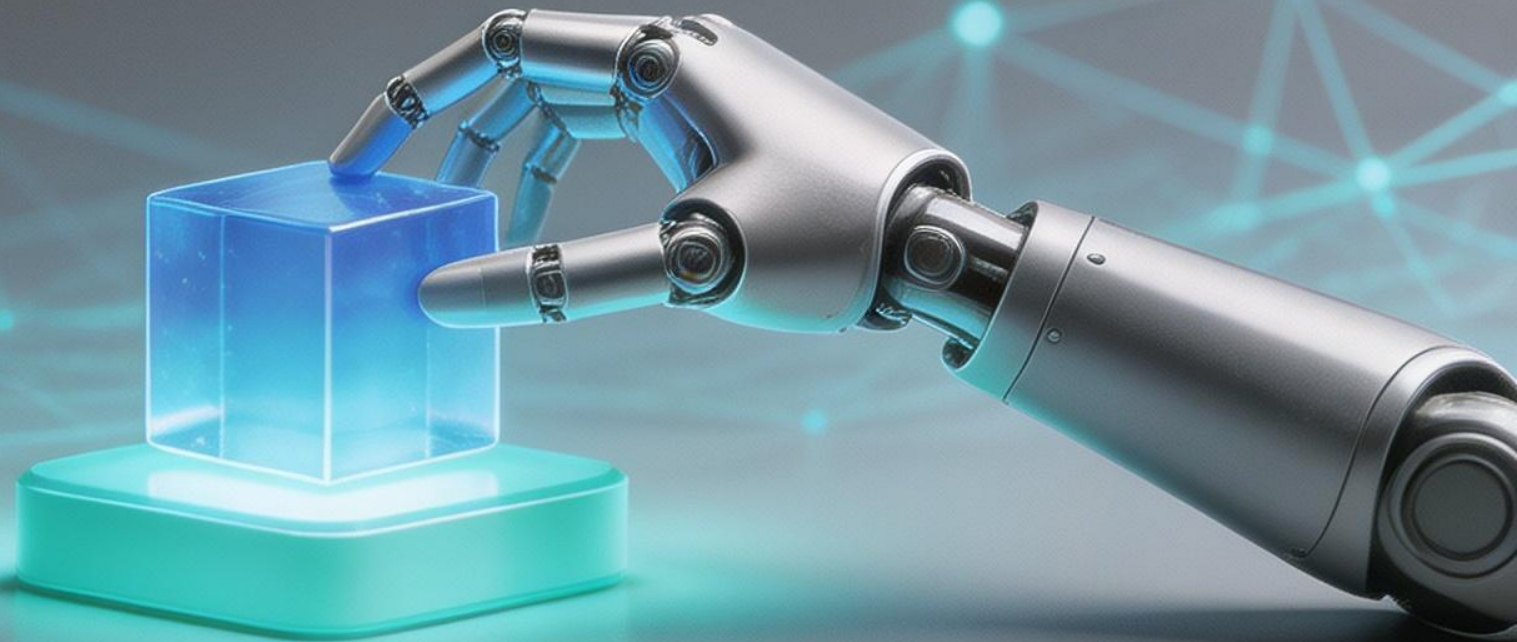
Fuses predictions from adjacent time steps to effectively suppress jitter and yield smoother motions.

Thanks!

Presenter: Mengyuan Wang

Date: 2026/08/15

Next: From VLM to VLA.





AVC Lab

Affective Vision Computing Lab

From VLM to VLA: Concepts, Architectures, Data & Training Methods

Presenter: Mengyuan Wang

Date: 2026/08/22

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01 **Define Embodied AI, VLM, VLA**

02 **Dissect the Architectures**

03 **Understand the Training Data**

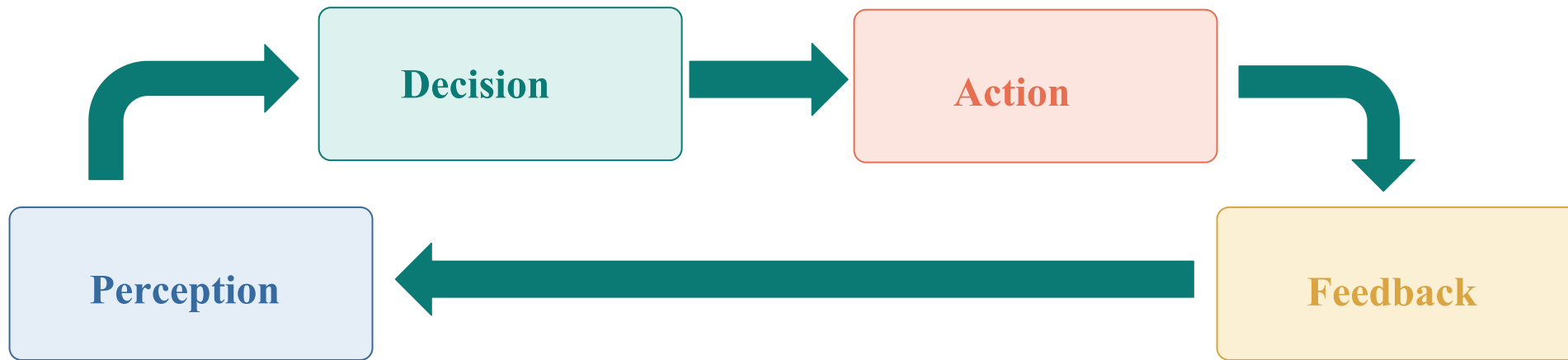
04 **Understand the Training Methods**

01

Define Embodied AI, VLM, VLA

What is Embodied AI?

- ❑ Embodied AI must possess a **physical embodiment**.
- ❑ It actively **interacts with the physical environment**.
- ❑ Distinct from Conversational AI and video-generating Generative AI.
- ❑ Core: Building a complete **Perception-Decision-Action-Feedback** physical loop.

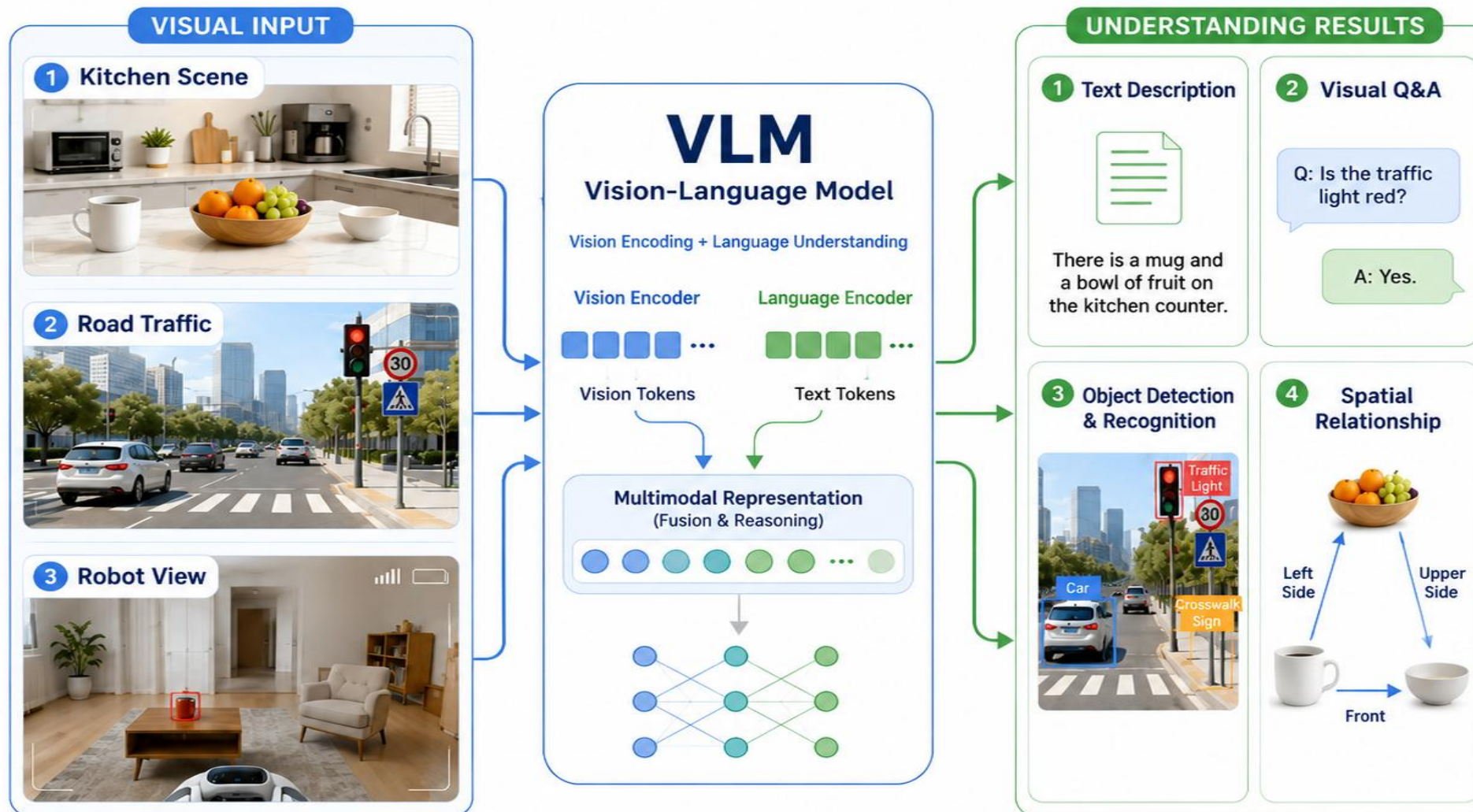


Some examples of VLM



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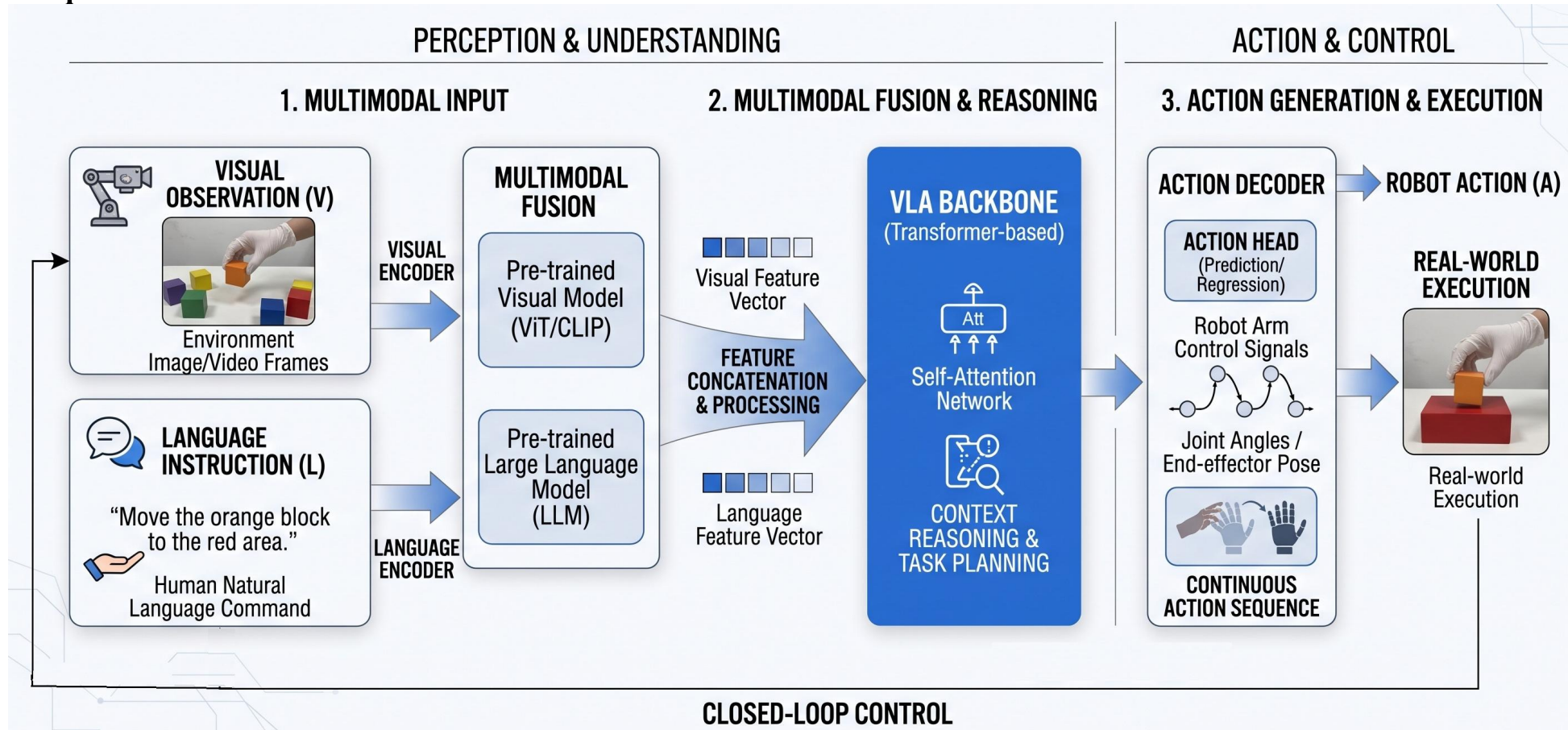


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What is VLA?

- ❑ Input: visual observations & natural language instructions & sensory modalities
- ❑ Output: control commands

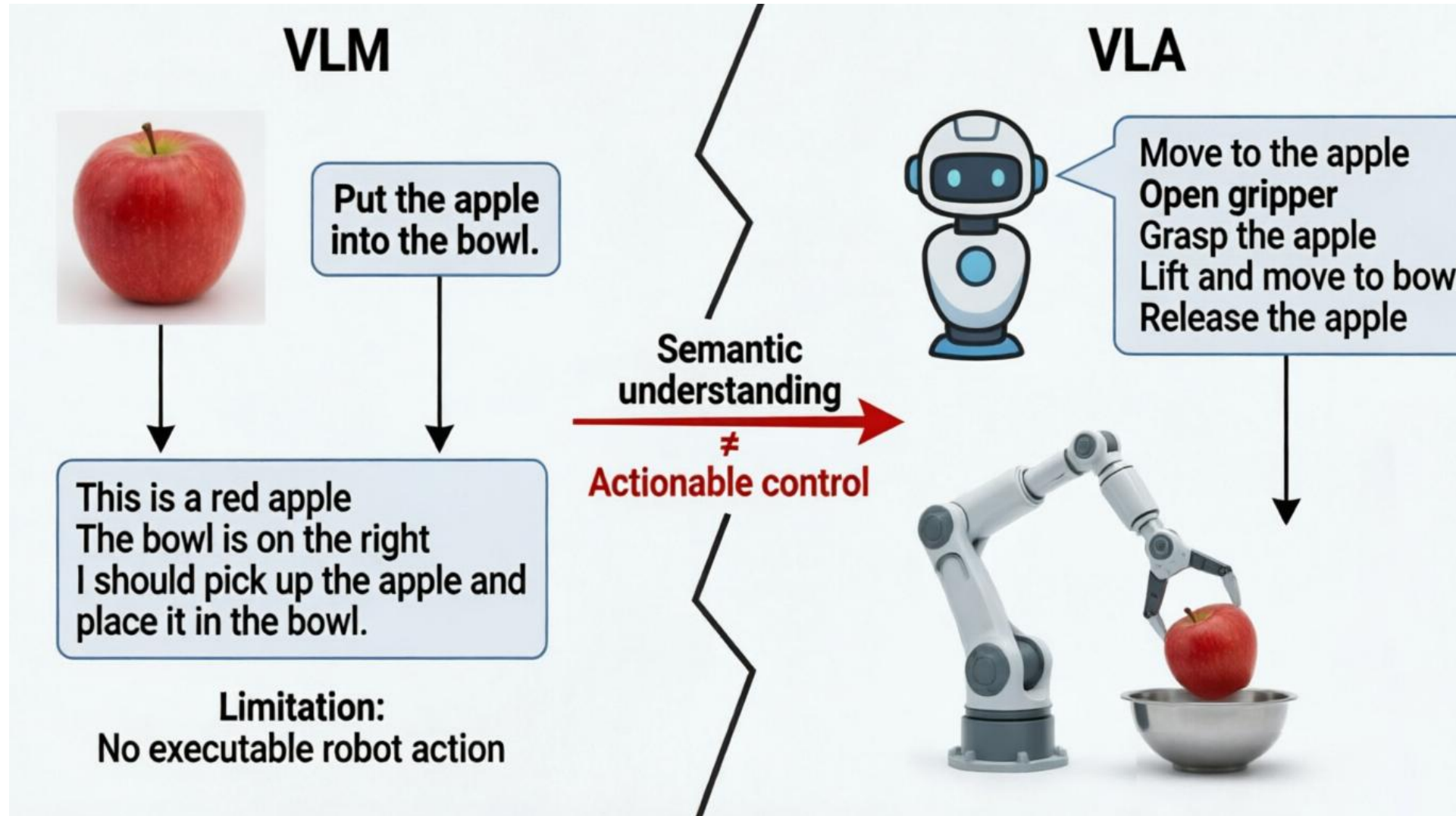


Why Not Directly Use VLMs?



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02

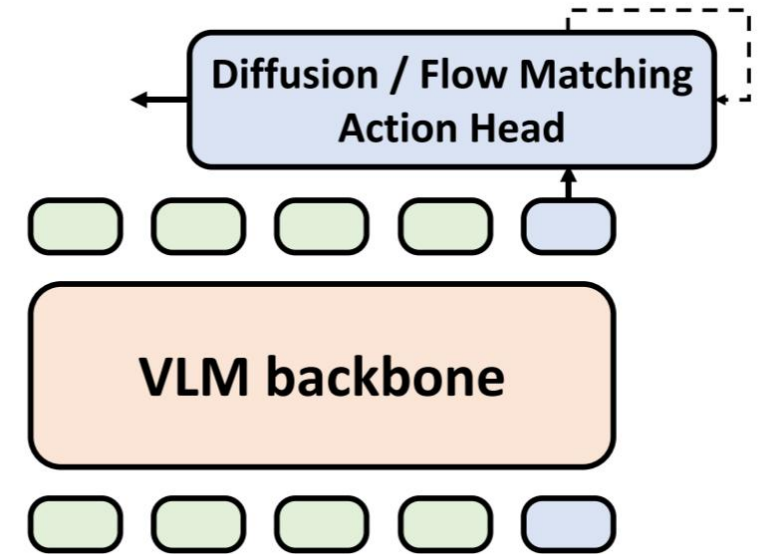
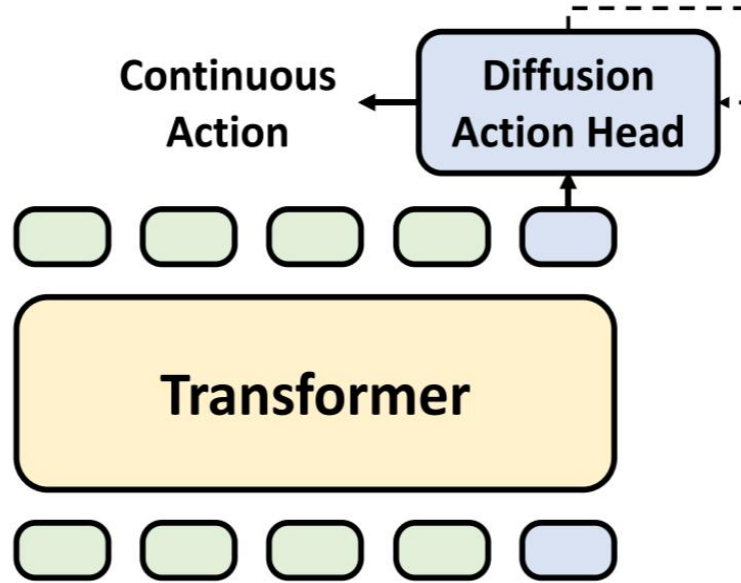
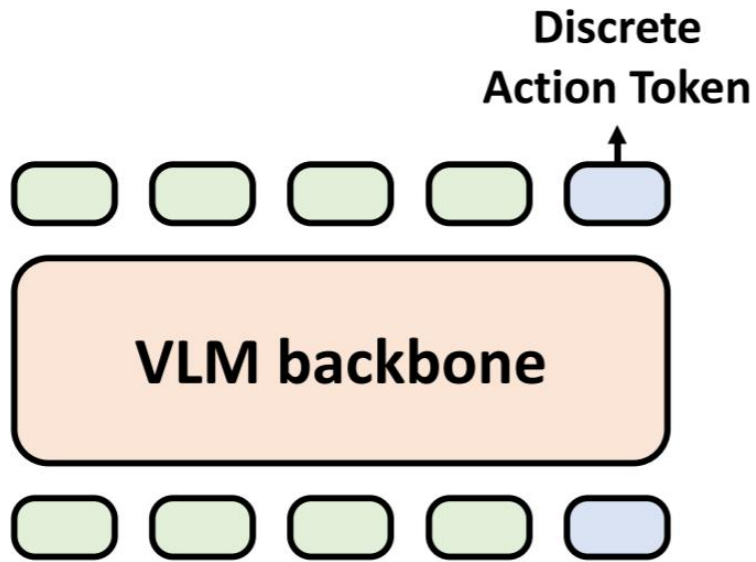
Dissect the Architectures

Three Core VLA Architectures



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(1) VLM + Discrete Action Token. (2) Transformer + Diffusion Action Head. (3) VLM + Flow Matching Action Head.

e.g., OpenVLA

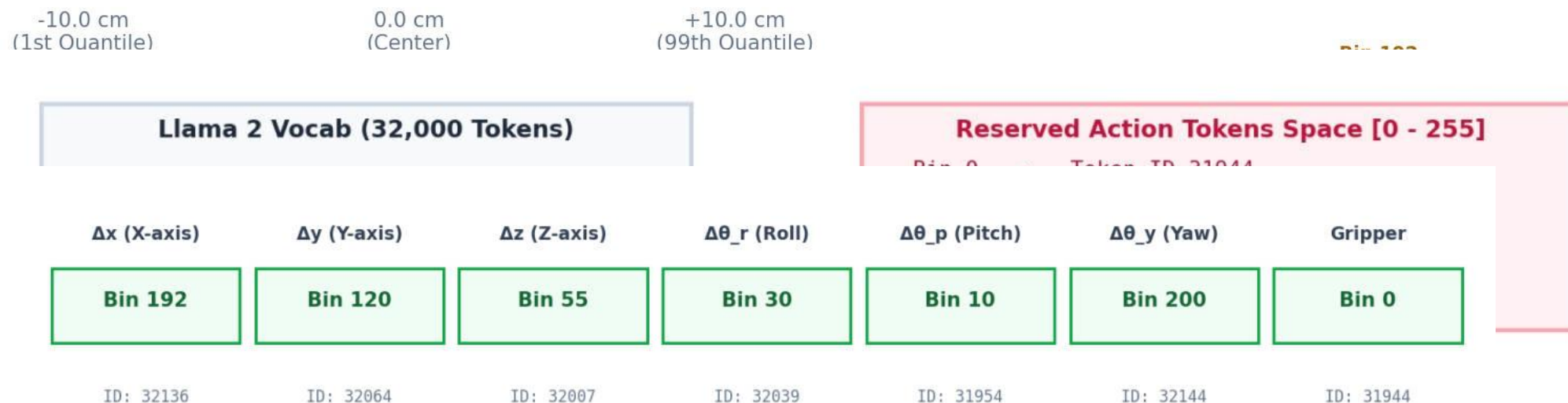
e.g., Octo

e.g., π_0

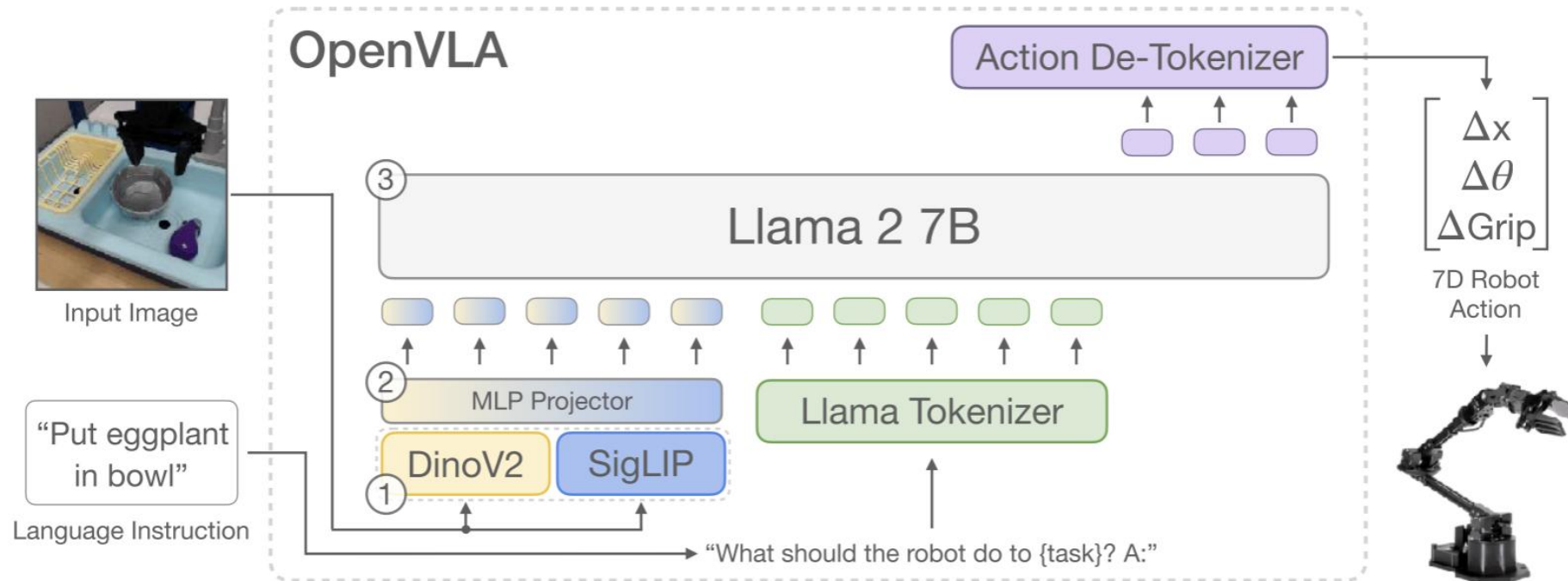
Preliminaries: discretized action tokens obtained via binning

- ❑ Discretize each dimension of the action space into bins (**typically 256**).
- ❑ Each bin ID is treated as a discrete token.
- ❑ Detailed Steps:

① Define Action Boundaries & Discretize into Bins

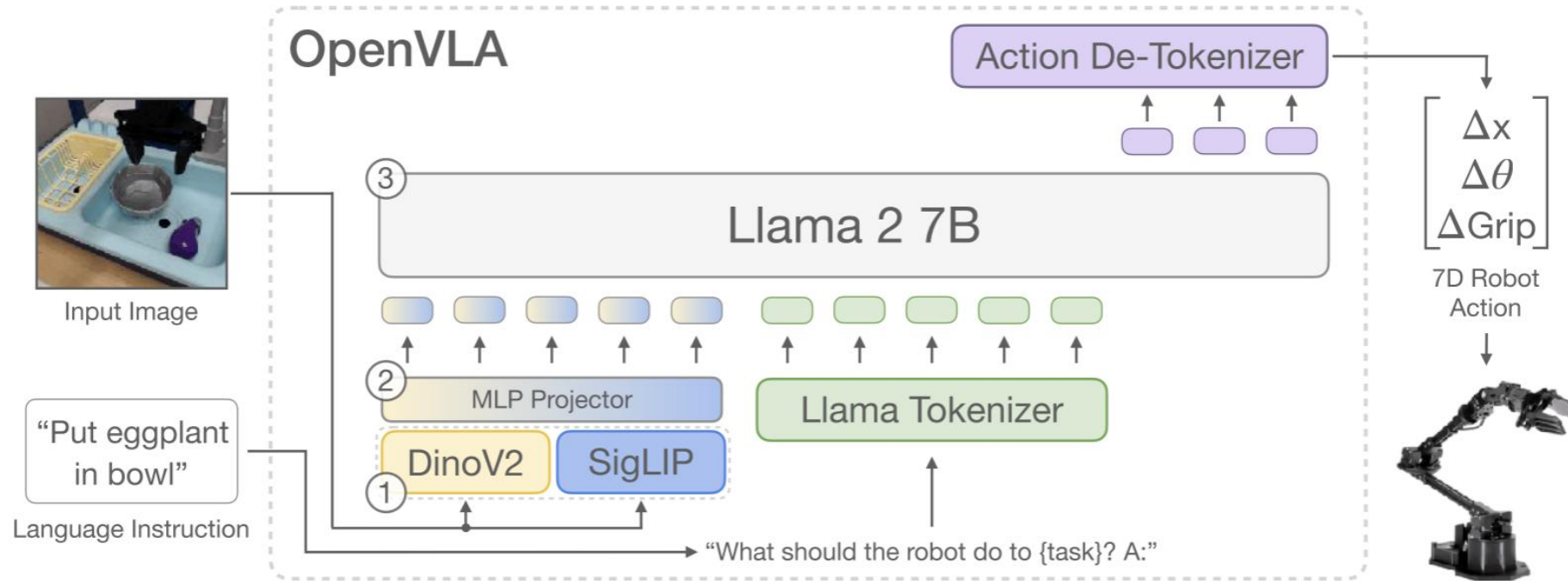


VLM + Discrete Action Token (OpenVLA)



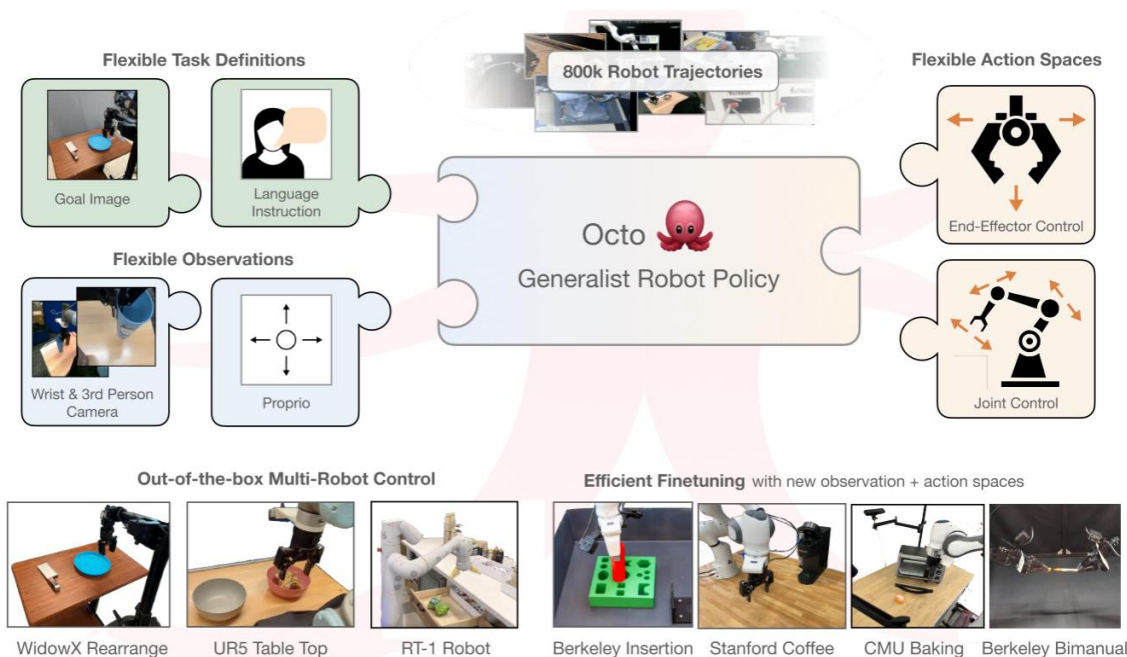
- ❑ Parameters: 7B
- ❑ Training Data: 970k real-world robot demonstrations (Open X-Embodiment)
- ❑ Full-parameter training: Unfreeze all parameters in OpenVLA
- ❑ Training Object & Loss: Next-token prediction with action-only cross-entropy loss

VLM + Discrete Action Token (OpenVLA)



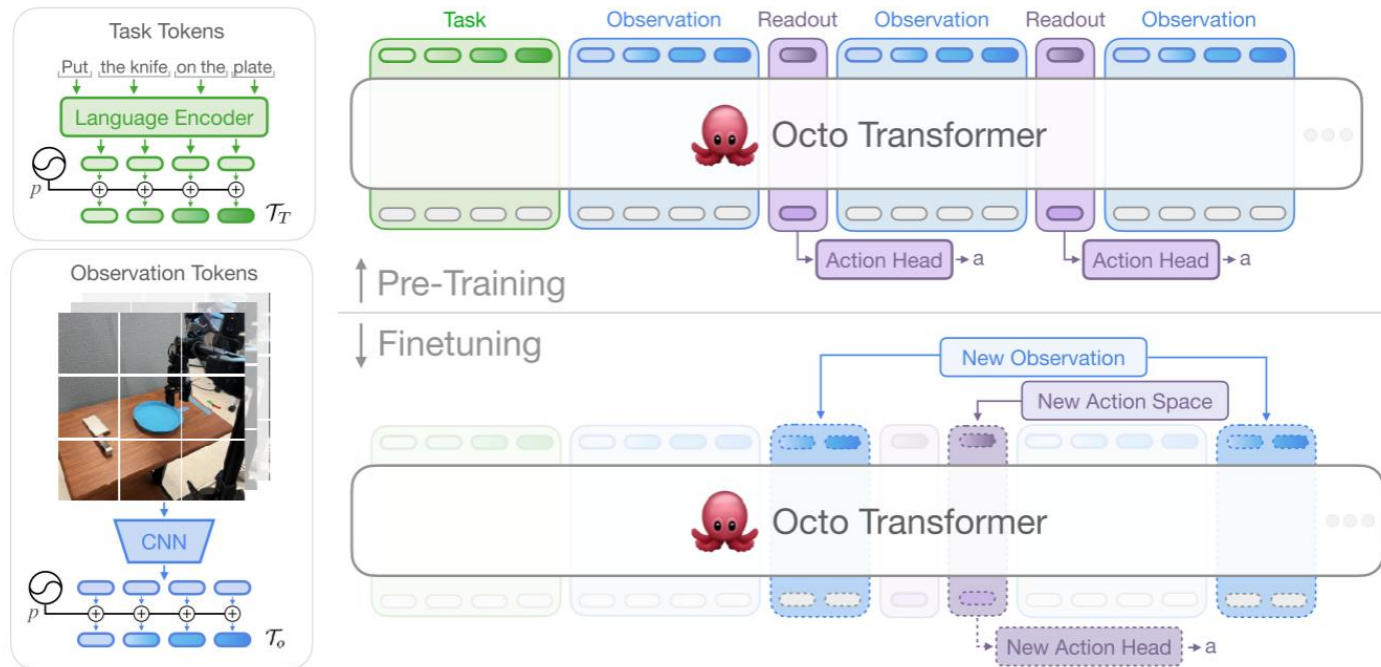
- ❑ Visual Encoder: 600M SigLIP (High-level semantic features) + Dino V2 (low-level spatial geometric details), Concatenated channel-wise
- ❑ Language Model Backbone: 7B Llama 2
- ❑ MLP projector: 2-layer
- ❑ Self-Attention Mechanism

Transformer + Diffusion Action Head (Octo)



- ❑ Training Data: 800K robot episodes (Open X-Embodiment)
- ❑ Flexible Observations
- ❑ Flexible Tasks
- ❑ Flexible Actions
- ❑ Parameters: Octo-Small (27M) / Octo-Base (93M)

Transformer + Diffusion Action Head (Octo)



❑ Causal Attention & Block-wise Attention

❑ Task Block: Bidirectional Attention

❑ Observation Block: $\mathcal{T}_{o,t}$ only attends to $\mathcal{T}_{o,0:t}$ & $\mathcal{T}_T$

❑ Read Tokens: $\mathcal{T}_{R,t}$ attends to $\mathcal{T}_o$ & $\mathcal{T}_T$ before it in the sequence (compact vector embedding of the observation sequence)

❑ Text Encoder: t5-base (111M)

❑ Visual Encoder: a lightweight CNN

❑ Input Tokenizers: $\ell, g, (o_1, \dots, o_H) \rightarrow [\mathcal{T}_\ell, \mathcal{T}_g, \mathcal{T}_o]$

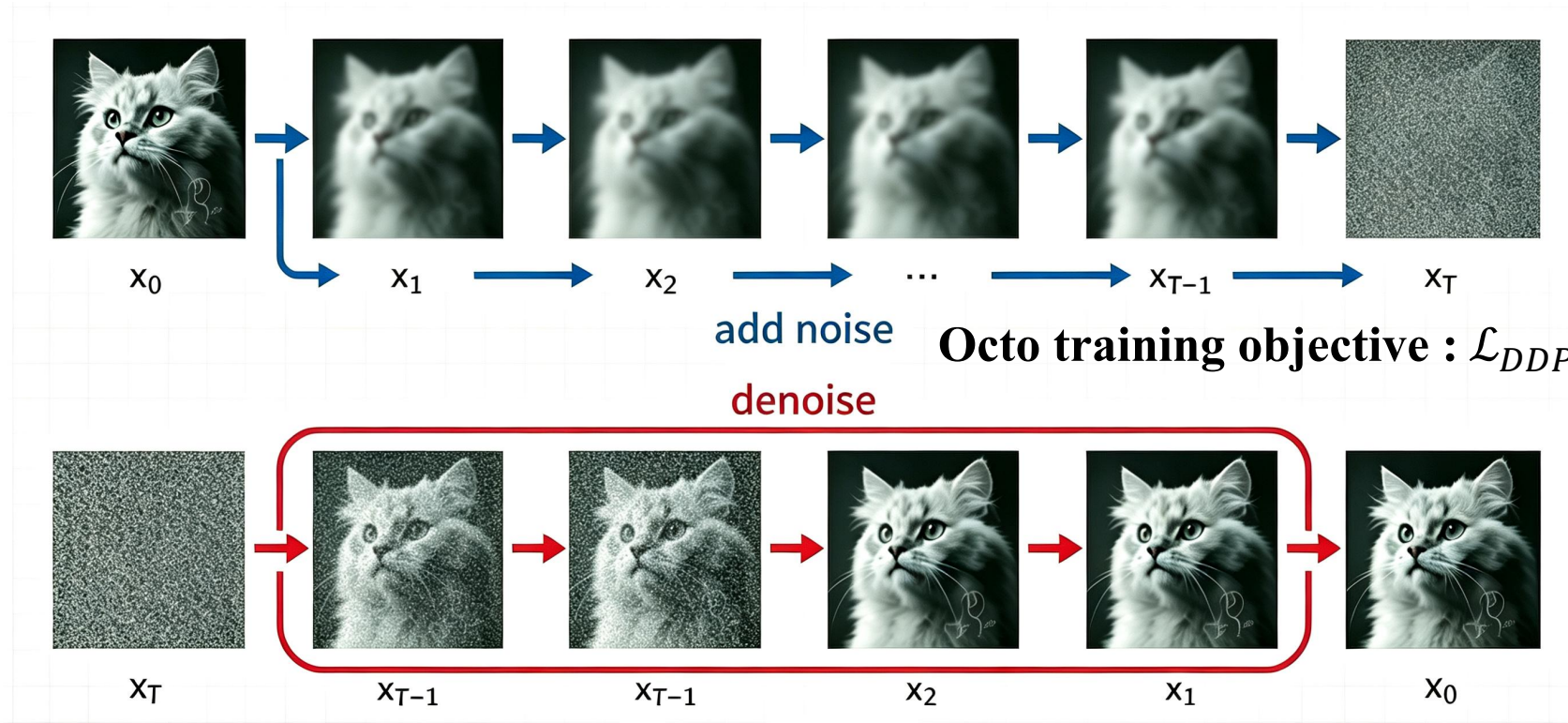
❑ Learnable position embeddings p

❑ Transformer Backbone:
 $e_\ell, e_g, e_o = T(\mathcal{T}_\ell, \mathcal{T}_g, \mathcal{T}_o)$

❑ Readout Heads: $R(e) \rightarrow a$

Transformer + Diffusion Action Head (Octo)

✦ Training objective (DDPM)

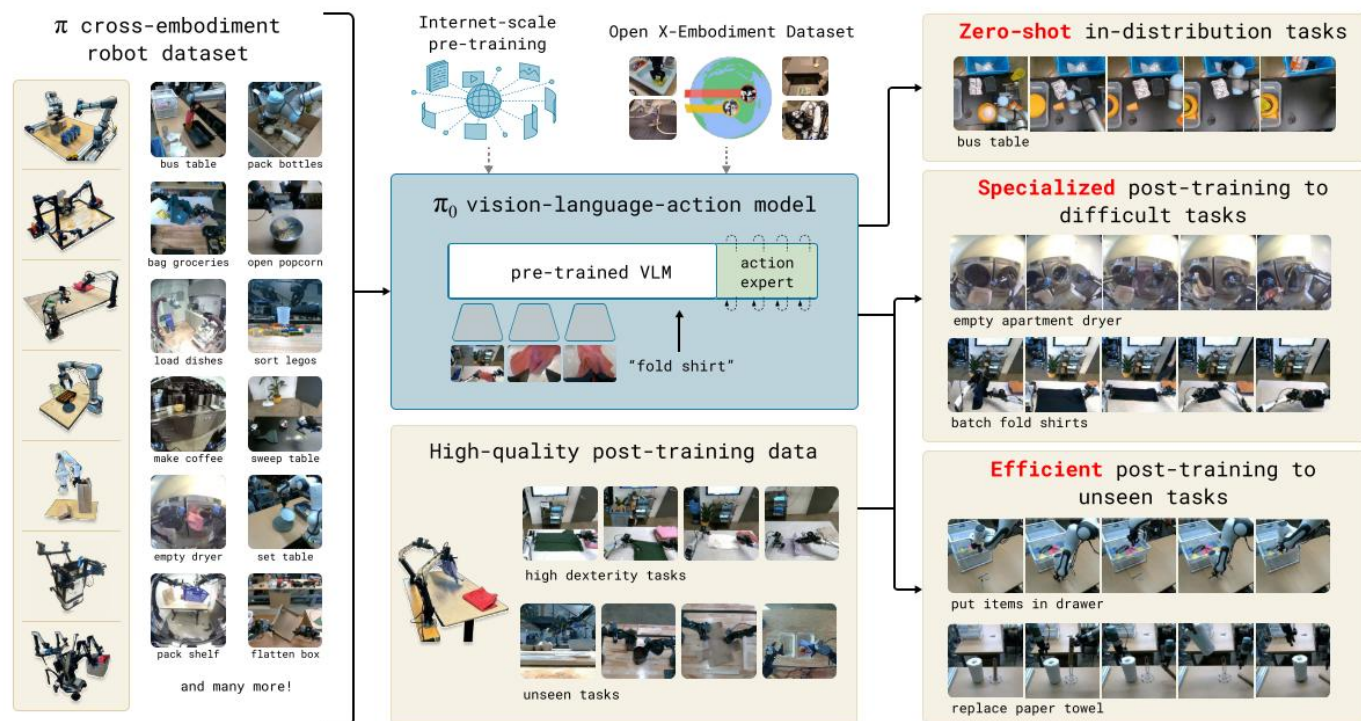


Octo training objective : $\mathcal{L}_{DDPM} = \mathbb{E}_{t,\epsilon,k} [\|\epsilon - \epsilon_\theta(a_t^k, e, k)\|^2]$

DDPM training objective: $L_{simple}(\theta) := \mathbb{E}_{t,x_0,\epsilon} [\|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t)\|^2]$

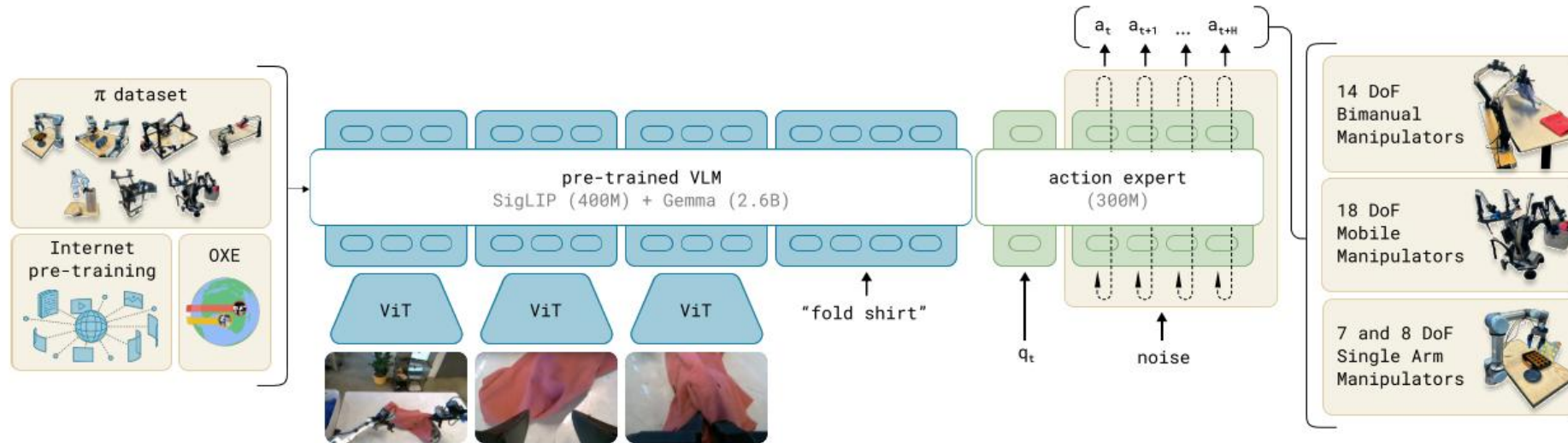
Denoising: $x^{k-1} = \alpha(x^k - \gamma \epsilon_\theta(x^k, e, k) + \mathcal{N}(0, \sigma^2 I)) \quad x^k \sim \mathcal{N}(0, I)$

VLM + Flow Matching Action Head (π_0)



- ❑ Parameters: 3.3B
- ❑ Training Data: 10,000 hours of dexterous manipulation data + open-source data
- ❑ Control Frequency: 50 Hz
- ❑ Training Paradigm: pre-training + post-training

VLM + Flow Matching Action Head (π_0)



□ Base Model: 3B PaliGemma (SigCLP + Gemma) + 300M Action expert

□ Input: $o_t = [I_t^1, \dots, I_t^n, \ell_t, q_t]$ $n = 2 \text{ or } 3$

□ Output: $A_t = [a_t, a_{t+1}, \dots, a_{t+H-1}] \sim p(A_t | o_t)$ $H = 50$

□ $[I_t^1, \dots, I_t^n, \ell_t] \rightarrow \text{VLM}$ $[q_t, A_t^\tau] \rightarrow \text{Action expert}$



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VLM + Flow Matching Action Head (π_0)

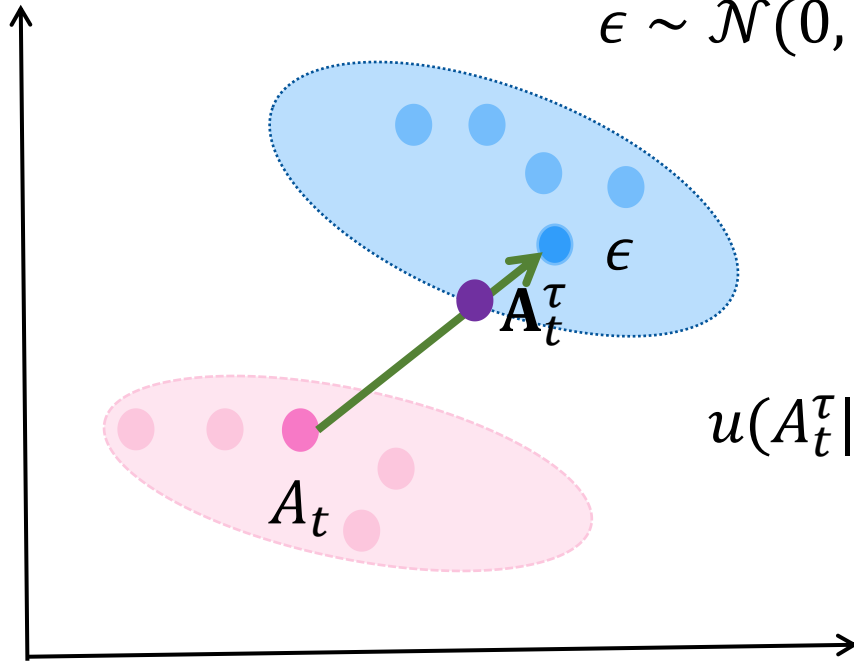
✦ Training objective (Flow Matching)

$$\epsilon \sim \mathcal{N}(0, I)$$

$$q(A_t^\tau | A_t) = \mathcal{N}(\tau A_t, (1 - \tau)I)$$

$$A_t^\tau = \tau A_t + (1 - \tau)\epsilon \quad \tau \in [0, 1]$$

$$u(A_t^\tau | A_t) = \epsilon - A_t \quad A_t = [a_t, a_{t+1}, \dots, a_{t+H-1}]$$



$$L^\tau(\theta) = \mathbb{E}_{p(A_t|o_t), q(A_t^\tau|A_t)} ||v_\theta(A_t^\tau, o_t) - u(A_t^\tau|A_t)||^2$$

$$\textit{Inference} : A_t^{\tau+\delta} = A_t^\tau + \delta v_\theta(A_t^\tau, o_t)$$

$$A_t^0 \sim \mathcal{N}(0, I)$$

$$\tau = 0 \rightarrow \tau = 1$$

03

Understand the Training Data

Datasets for VLA



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1. HUMAN DATASETS

Ego4D

EPIC-KITCHENS

ARCTIC

OAKINK2

EMBODIMENT GAP

2. SIMULATION DATASETS

RoboTurk

MimicGen

DexMimicGen

SIM-TO-REAL GAP

3. REAL ROBOT DATASETS

QT-Opt

BridgeData V2

DROID

Open X-Embodiment (OXE)

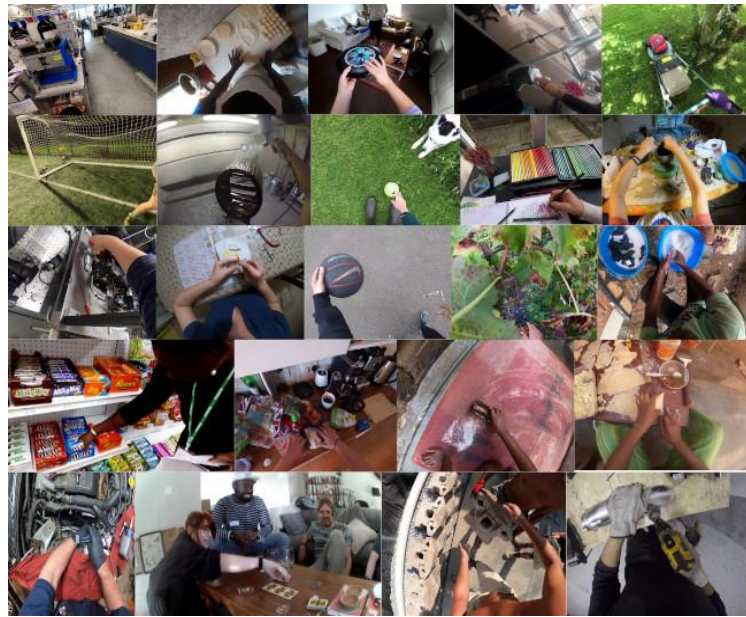
SCARCITY & HETEROGENEITY

Datasets for VLA- Some examples

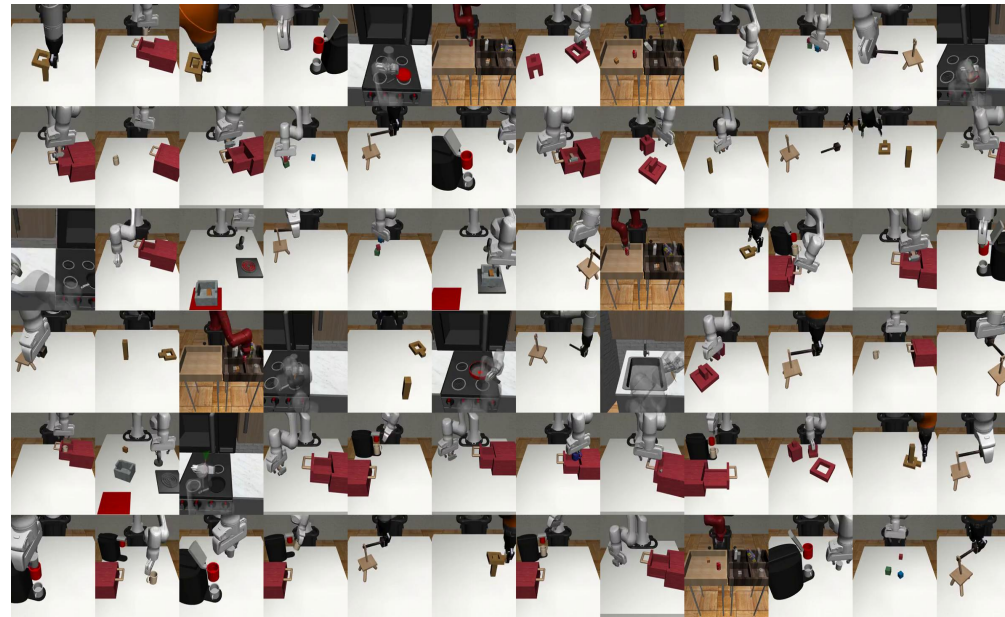


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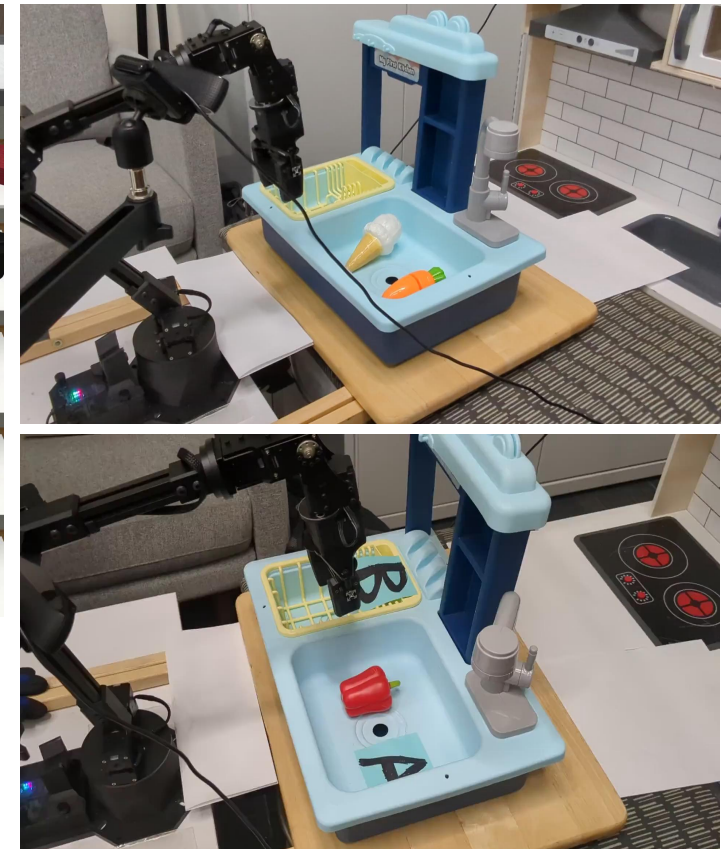
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❑ Human datasets (Ego4D)



❑ Simulation datasets (MimicGen)



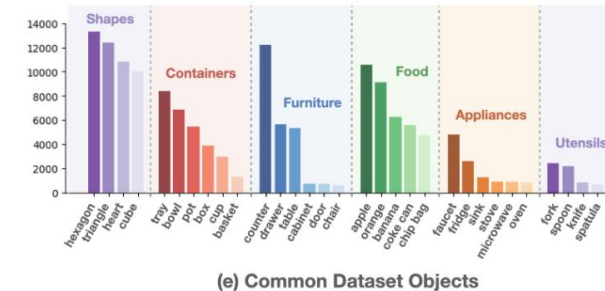
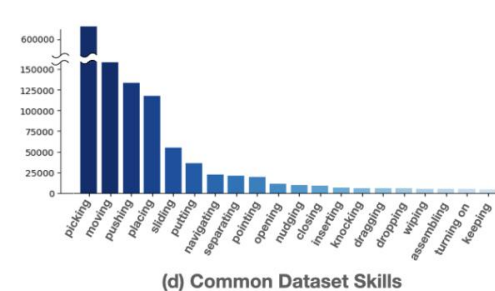
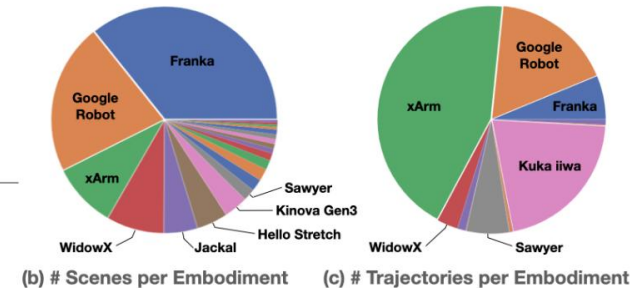
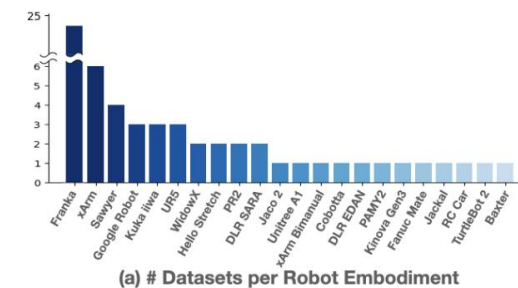
❑ Real robot datasets (OXE)

Open X-Embodiment: The ImageNet Moment for VLA



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- ❑ 22 robot embodiments
- ❑ 1M+ real robot trajectories

- ❑ 527 skills (160266 tasks)
- ❑ 34 robotic research labs

04

Understand the Training Methods

Two-Stage Pipeline



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□ Pre-training

- Acquire **general capabilities** and promote **transferability across diverse robotic embodiments**.
- Robot demonstrations + Web-scale vision-language corpora + Structured annotations
→ Improve semantic understanding and visuomotor grounding
- Initialized from VLM backbones

□ Post-training

- Refined using **high-quality robot demonstration data** to improve performance on **specific downstream tasks**.
- Requires high-quality, robot and task-specific data
- Full Fine-Tuning vs. LoRA Fine-Tuning vs. Action-Head-Only Fine-Tuning

References



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- ✓ Ma, Yuen, et al. "A survey on vision–language–action models for embodied ai." IEEE Transactions on Neural Networks and Learning Systems (2026).
- ✓ Kawaharazuka, Kento, et al. "Vision-language-action models for robotics: A review towards real-world applications." IEEE Access (2025).



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World Models & World Action Models

Presenter: Mengyuan Wang

Date: 2026/08/29

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- 02** **Two Tech Routes for World Models**

- 03** **Architecture of World Action Models**

- 04** **Data & Evaluation**

01

From VLA to World Models

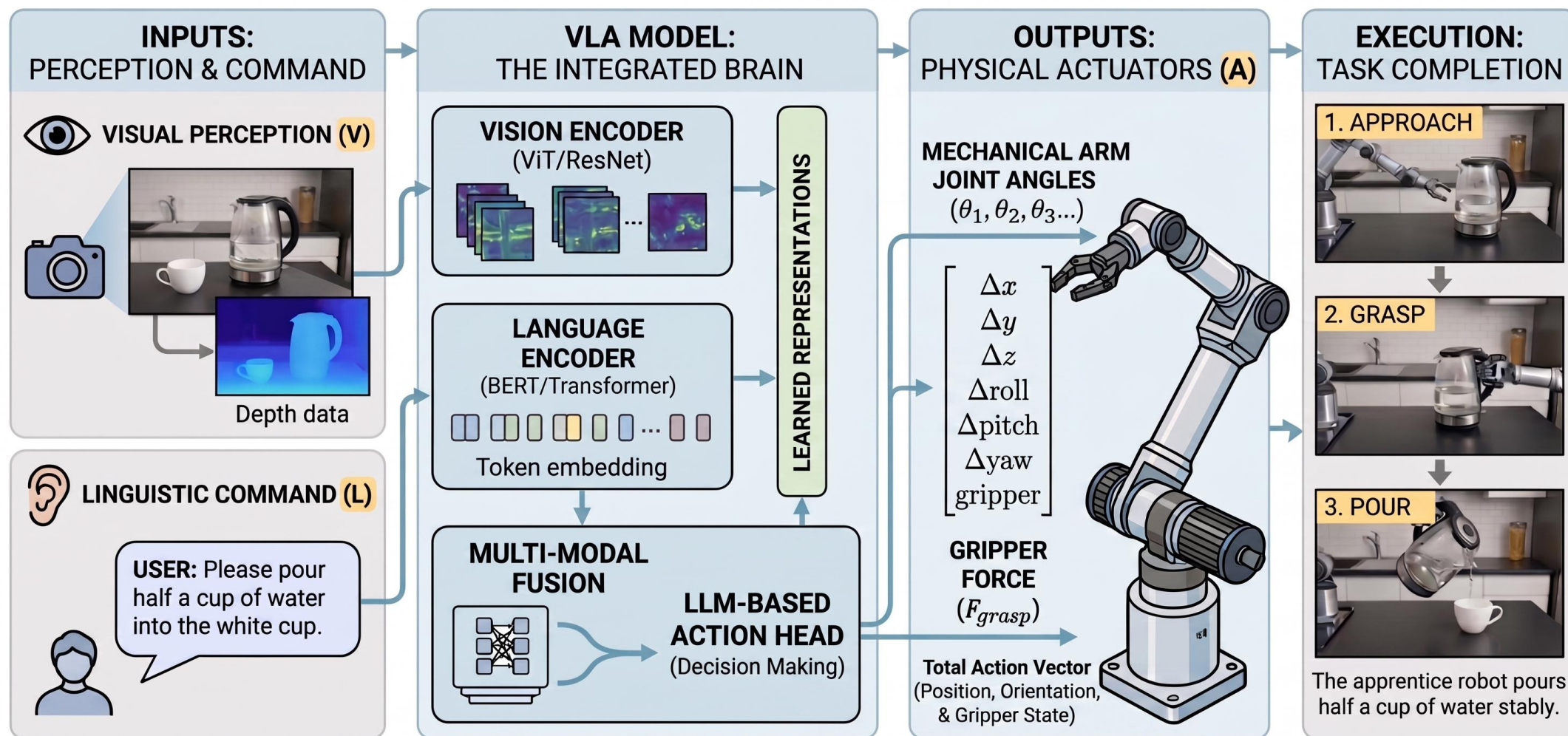
Recap: Vision-Language-Action



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□ **VLA:** Observation o + Linguistic Instructions $l \rightarrow [a_1, a_2, \dots, a_T]$

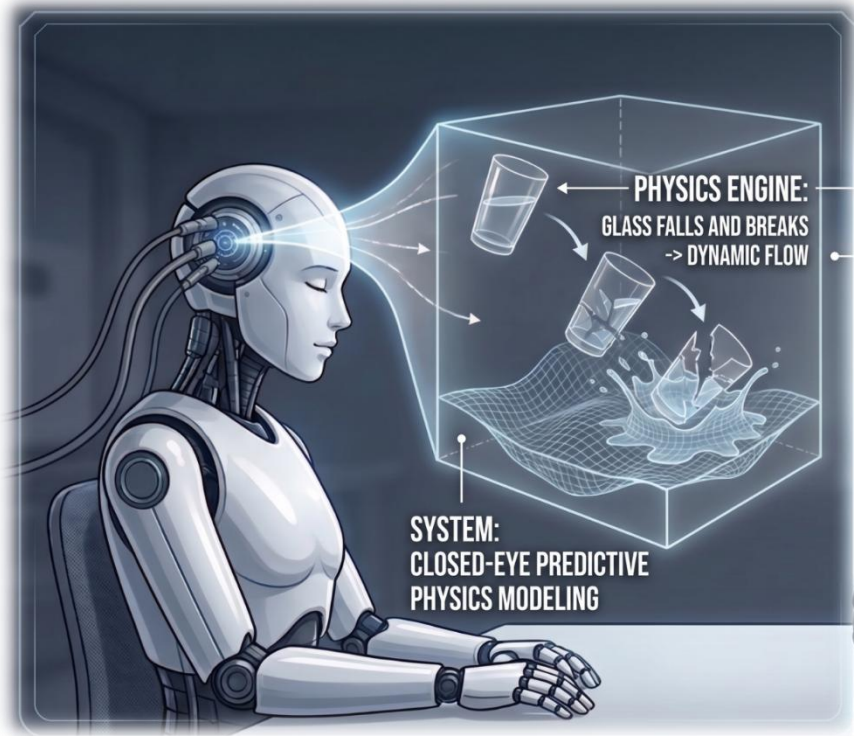




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What is World Models?



❑ **State-transition model:** current state & action $\rightarrow$ next state or a sequence of future states

❑ **Two directions (for robotic policies):**

✦ **Tighter Coupling:** predictive modeling + action generation. **WAM**

✦ **Simulators:** Using world models as simulators for validation, post-training, and RL.

❑ **Equation:** $\mathcal{L}_{WM} = \mathbb{E}_{(o,a,o') \sim \mathcal{D}} [-\log p(o'|o, a)]$

❑ **Three core capabilities of actionable world model**

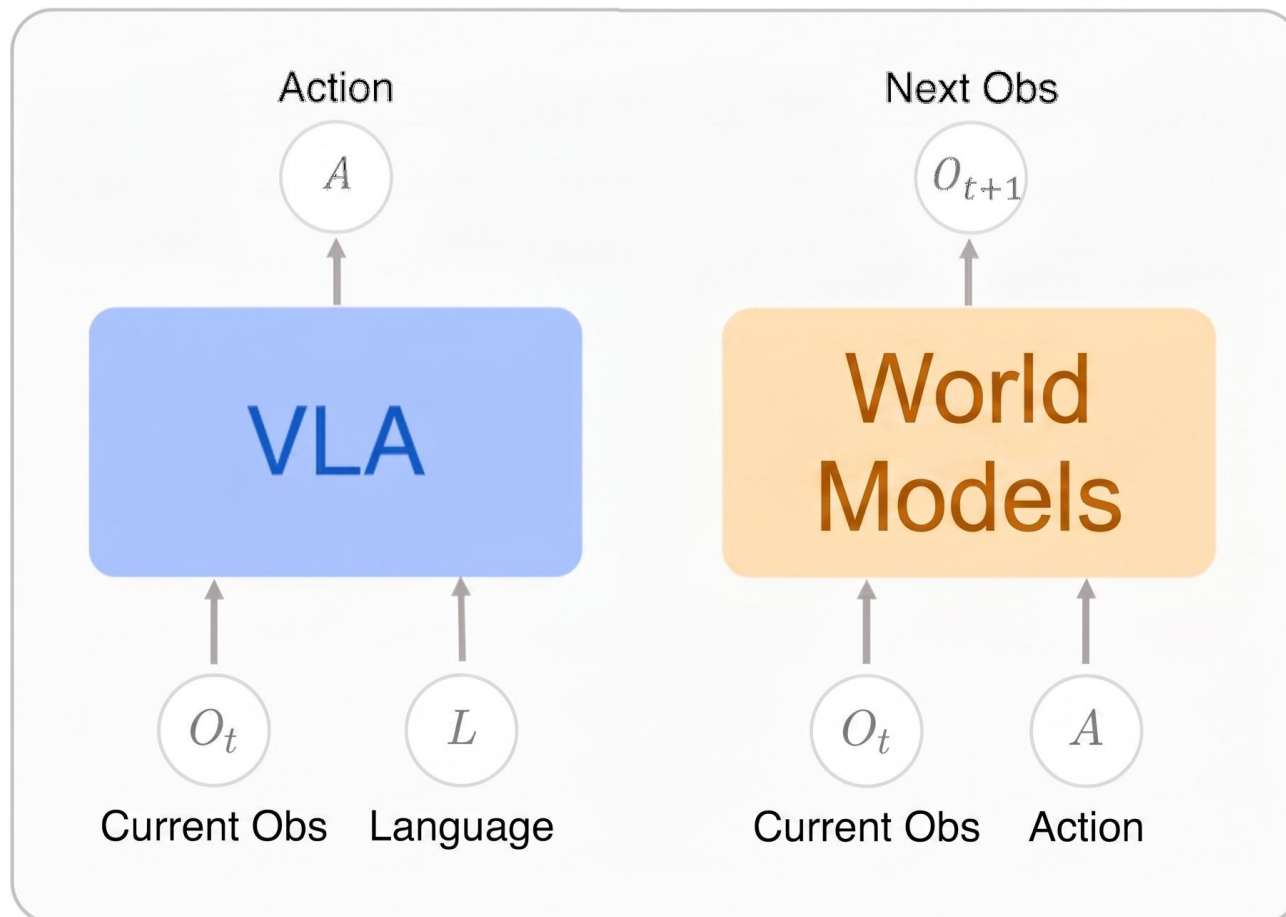
- ✦ Foresight
- ✦ Imagination-driven planning
- ✦ Data amplification

World Models vs. VLA



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□ Presentation Format

□ $\mathcal{L}_{VLA} = \mathbb{E}_{(o,l,a) \sim \mathcal{D}}[-\log p(a|o,l)]$

□ $\mathcal{L}_{WM} = \mathbb{E}_{(o,a,o') \sim \mathcal{D}}[-\log p(o'|o,a)]$

□ Focus

□ “What should I do now?”

□ “If I do X, what happens next?”

AC-WMs & LC-WMs

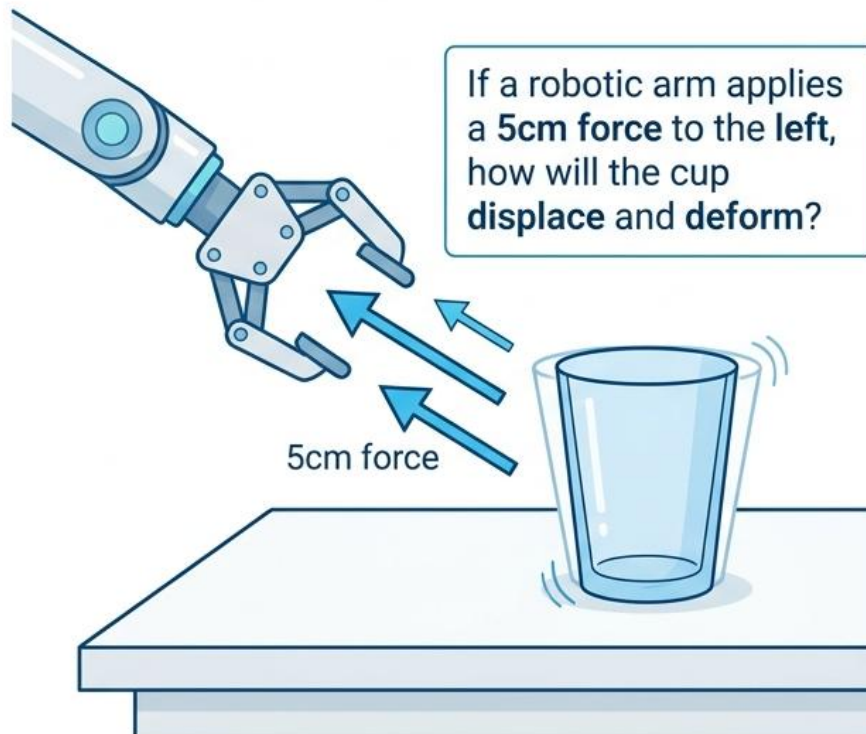


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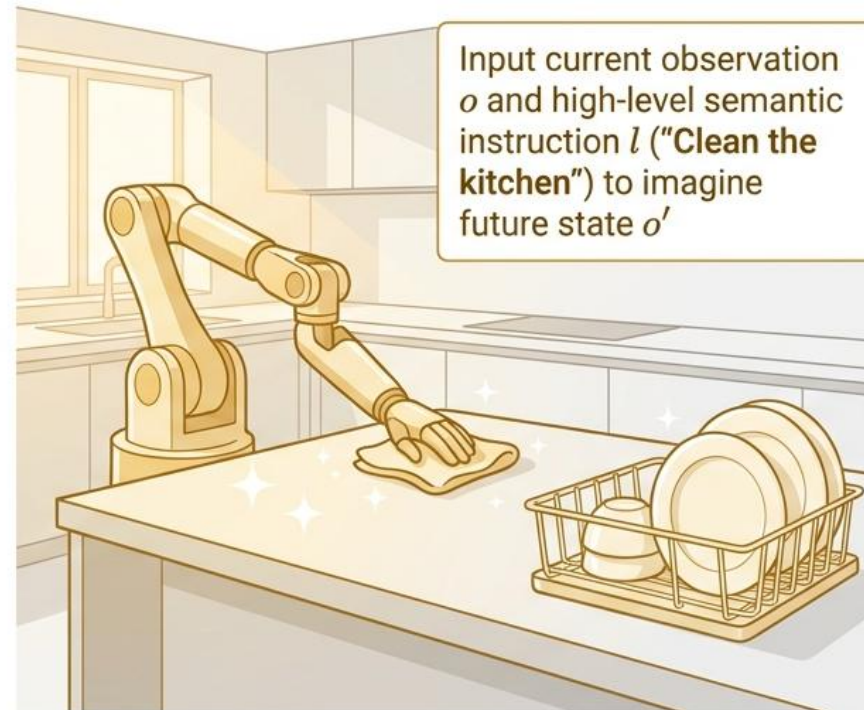
ACTION-CONDITIONED

$$P(o_{t+1} \mid o_t, a_t)$$



LANGUAGE-CONDITIONED

$$P(o_{t+1} \mid o_t, l_t)$$



Why do robots need world models?



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Dimensional Upgrade of Space

- ❑ **LLM:** 1D + t
- ❑ **Autonomous Driving:** 2D + t
- ❑ **Embodied AI Robots:** 3D + t
- ❑ **3D Forward Physics Sandbox:**
World models act as **internal dynamical systems**, accurately simulating 3D space, depth consistency, and rigid-body collisions in their "mind".



Open-World Generalization

- ❑ **OOD Generalization Failure:**
VLA generalization is severely bottlenecked by data.
- ❑ **Bypassing Data Bottlenecks via Video Self-Supervision**
- ❑ **OOD Generalization via Spatiotemporal Priors**



Mitigating Compounding Errors

- ❑ **Reactive Mapping Bottleneck:**
long-horizon reasoning and temporal credit assignment extremely difficult
- ❑ **Compounding Error Avalanche**
- ❑ **World Models Foresight & Correction**



Feature-level Latent Prediction

- ❑ **Computational Waste of Pixel-Level Reconstruction**
- ❑ **Abstract Latent Space:** Predict physical state transitions and causal alignment.

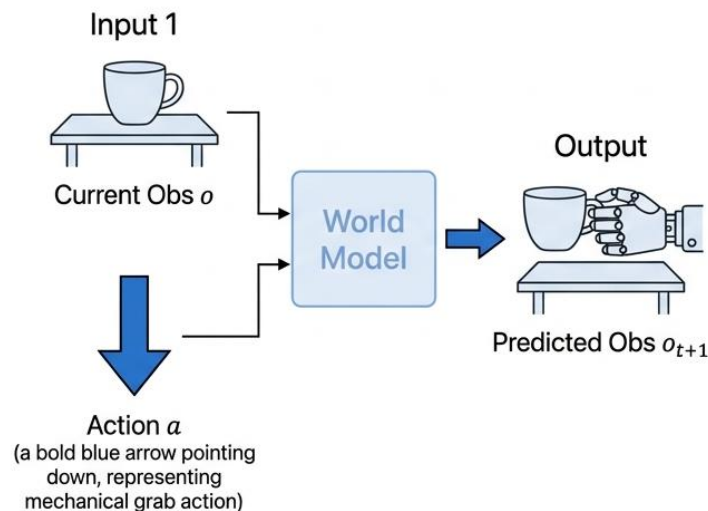
World Models $\neq$ Video Generation Models



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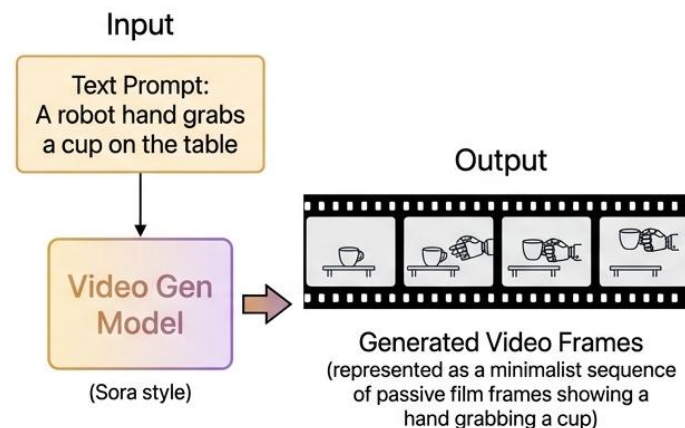
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World Model
(Action-conditioned prediction)



Interactive Foresight for Robot Planning

Video Generation Model
(Passive synthesis)



Passive Text-to-Video Synthesis



World Models

- Action-conditioned Commitment: $P(o'|o, a)$
- Pixels are merely optional proxies, not the objective.
- Explicit & Implicit/Latent-space Prediction



Video Generation Models

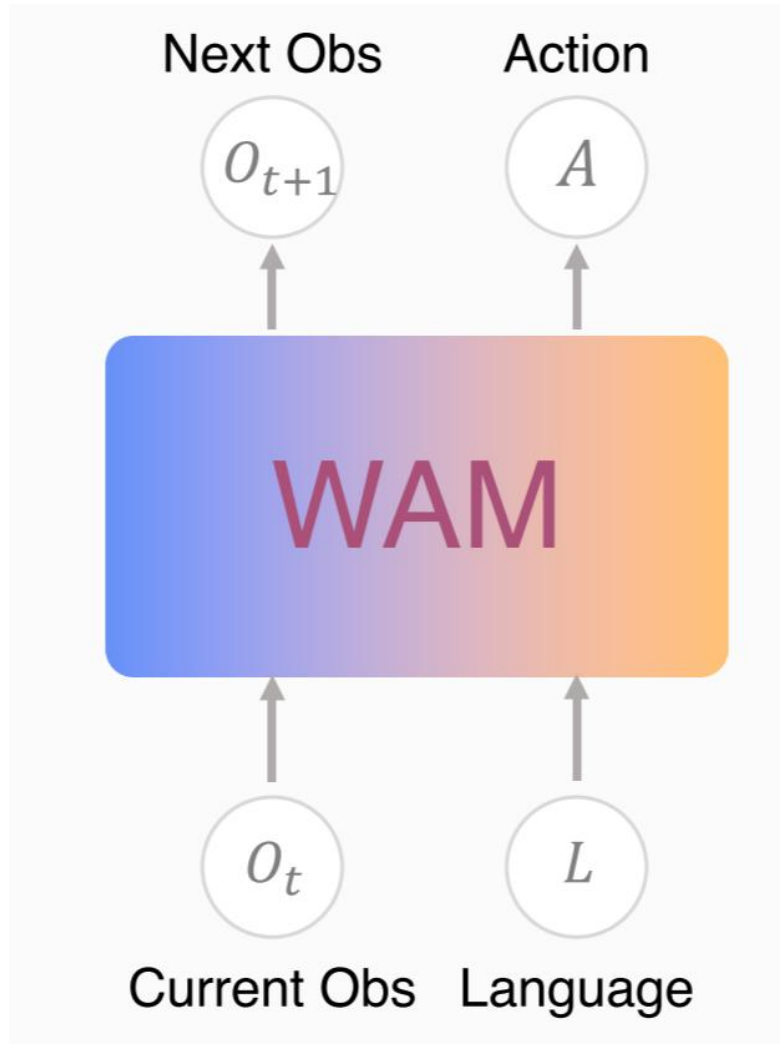
- Passive Visual Continuation: $P(v|o, l)$
- Pixel-level Reconstruction



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World Action Models



- **Presentation Format**

$$\mathcal{L}_{WAM} = \mathbb{E}_{(o,l,o',a) \sim \mathcal{D}} [-\log p(o', a | o, l)]$$

- **Environmental Dynamics Modeling (World Modeling) + Motor Control (Action Generation)**

- **Two Core Criteria**

- **Forward Predictive Modeling**

- Explicit Visual Predictions

- Implicit Physical Representations

- **Coupled Action Generation**

- Achieve deeper **physical understanding** and **stronger zero-shot generalization**.

World Action Models vs. Video Action Models



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Video generation is merely a means of physical inference, rather than the ultimate definition of a world model.

➤ **Superset vs. Subset**

- **WAMs:** A broader, modality-independent superset of predictive agents.
- **VAMs:** Aligning action with synthesized visual futures.

➤ **Multi-modal State vs. Pixel-Bound**

- **WAMs:** Single-image state transitions, Dense point clouds, Tactile, Force feedback
- **VAMs:** Pixel-space video generation

➤ **Causal Dynamics vs. Pixel Synthesis**

- **WAMs:** Underlying Physical Laws & Causal Dynamics
- **VAMs:** Generate visually rich and physically plausible action videos.

02

Two Tech Routes for World Models

How to represent the future state of world models?



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The core divergence lies in state representation: explicit methods generate RGB pixels, while implicit methods use compressed latent features.



Explicit World Models

- ❑ **Definition:** World models directly predict **future state transitions** within the **RGB image** or **video pixel space**.
- ❑ **Core:** Preserve rich spatial geometry, textures, and environmental details.
- ❑ **Limitation:**
 - ❑ Inference Latency Tax
 - ❑ Demand Massive Computational Resources

Explicit Pixel-level Prediction

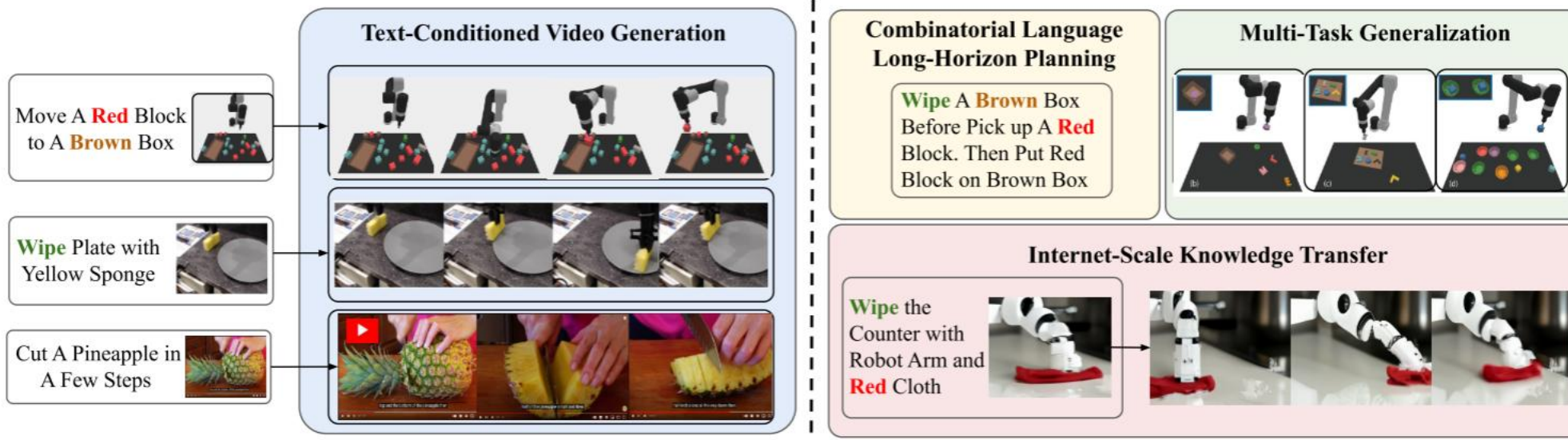


Implicit World Models

- ❑ **Definition:** Models completely abandon pixel rendering, instead predicting **physical state transitions** directly within an **abstract latent space**.
- ❑ **Core:** Predict the evolution of high-dimensional physical features via latent alignment.
- ❑ **Limitation:**
 - ❑ Lack of Interpretability
 - ❑ Potential Loss of Fine-grained Details

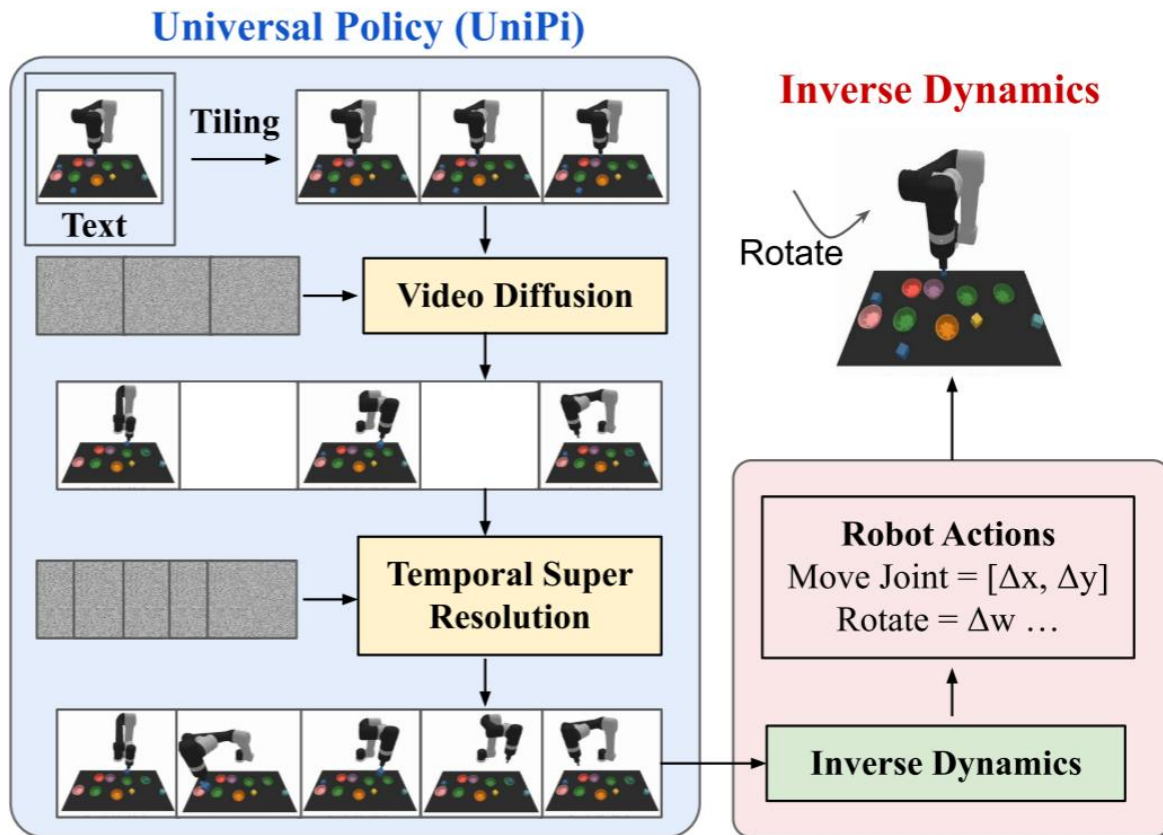
Implicit Latent-space Dynamics Models

Explicit Route: UniPi Case Study



- ❑ **Universal Policy:** Sequential Decision Making Problem → Text-Conditioned Video Generation Problem
- ❑ **Sources of Data:** Simulated, Real Robots, YouTube
- ❑ **Advantages:**
 - ❑ Combinatorial Generalization
 - ❑ Action Planning (Video Generation Procedure)
 - ❑ Multi-task Learning
 - ❑ Internet-Scale Knowledge Transfer

Explicit Route: UniPi Case Study



❑ Cascaded Predict-then-Act

❑ Video Diffusion Planner

$$\rho(\cdot | x_0, c): \mathcal{X} \times \mathcal{C} \rightarrow \Delta(\mathcal{X}^H)$$

❑ Inverse Dynamics Model

$$\pi(\cdot | \{x_h\}_{h=0}^H, c): \mathcal{X}^{H+1} \times \mathcal{C} \rightarrow \Delta(\mathcal{A}^H)$$

❑ Dataset: $\mathcal{D} = \left\{ \left((x_i, a_i)_{i=0}^{H-1}, x_H, c \right) \right\}_{j=1}^n$

❑ Pixel-Space: $\rho(\tau | x_0, c) \rightarrow \tau = [x_1, \dots, x_H] \in \mathcal{X}^H$

❑ Conditional Video Synthesis.

❑ Trajectory Consistency through Tiling.

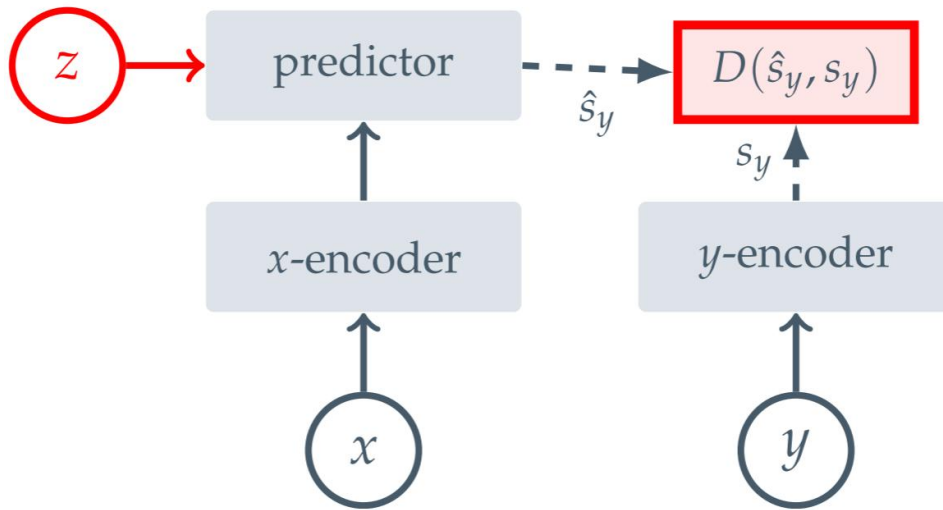
❑ Hierarchical Planning.

❑ Flexible Behavioral Modulation (Test-time Adaptability)

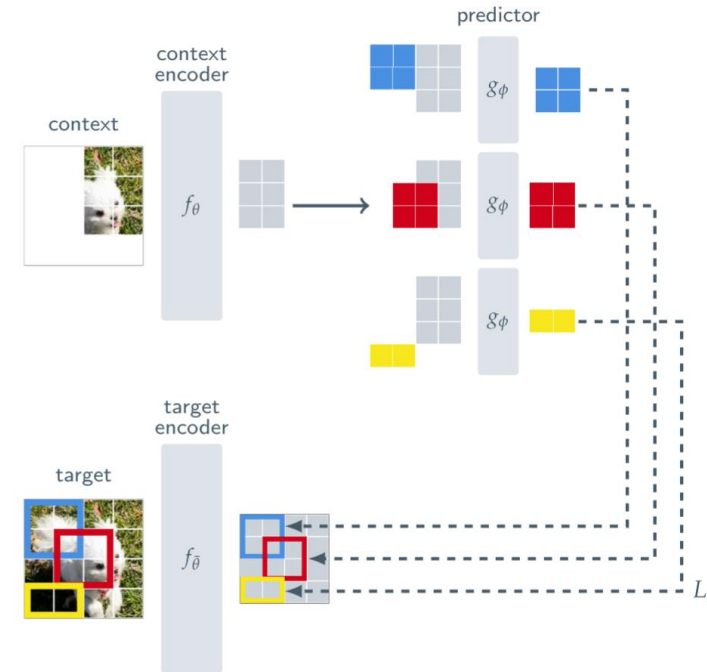
❑ Learned Classifier on Images

❑ Goal Image

Implicit Route: Predictive Representation Learning via I-JEPA



(a) Joint-Embedding Predictive Architecture



(b) Image-based Joint-Embedding Predictive Architecture

❑ **Asymmetric Architecture:** Context Encoder (f_θ) & Target Encoder ($f_{\bar{\theta}}$) & Predictor (g_ϕ) (ViT)

❑ **Targets:** Image $y \rightarrow$ Sequence of N Non-overlapping Patches $\rightarrow f_{\bar{\theta}} \rightarrow \mathbf{s}_y = \{\mathbf{s}_{y_1}, \dots, \mathbf{s}_{y_N}\} \rightarrow$ Randomly sample M blocks $\mathbf{s}_y(i) = \{\mathbf{s}_{y_j}\}_{j \in B_i}$ ($M = 4$) B_i : Mask corresponding of the i^{th} block

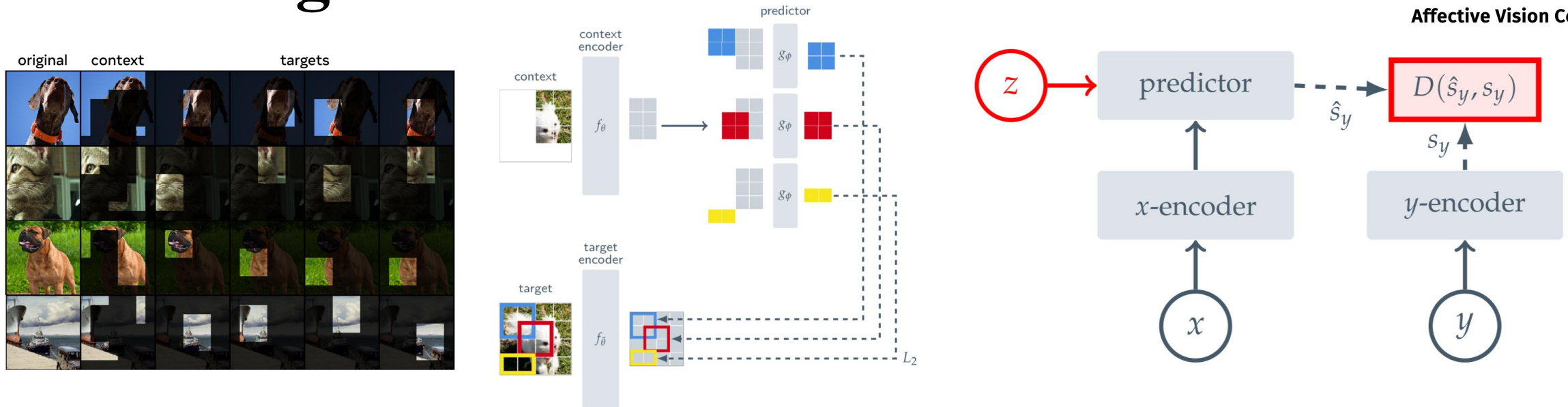
❑ **Context:** Sample a Single Block x (1:1) $\rightarrow$ Remove any overlapping regions from the context block $\rightarrow f_\theta \rightarrow$ Patch-level representation $\mathbf{s}_x = \{\mathbf{s}_{x_j}\}_{j \in B_x}$

Implicit Route: Predictive Representation Learning via I-JEPA



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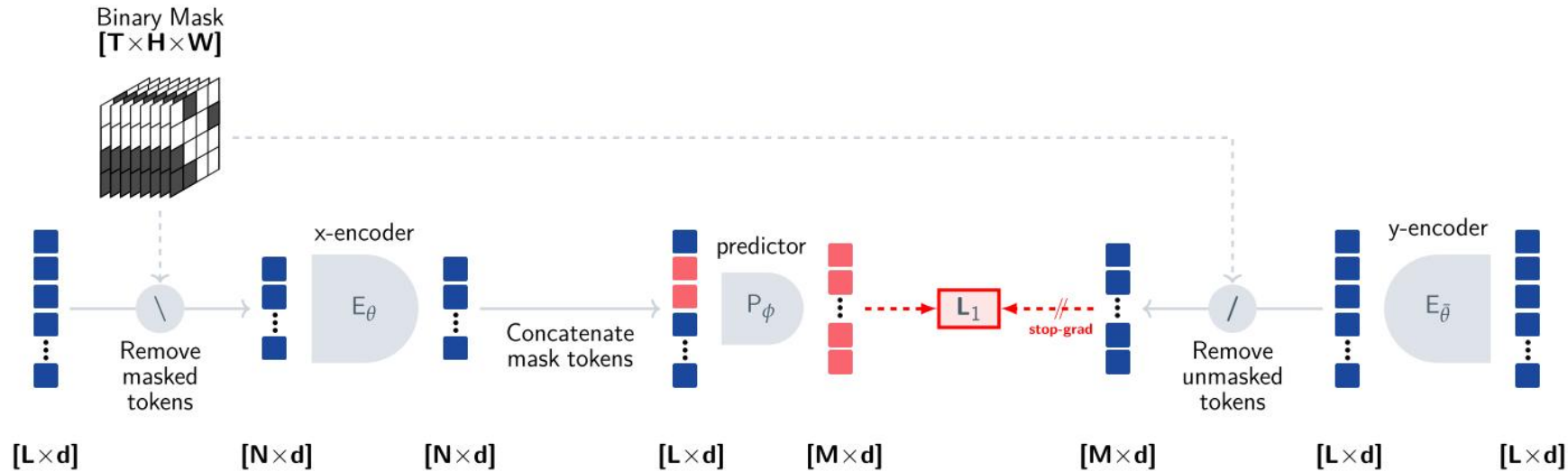
□ **Prediction:** $s_x \rightarrow g_\phi(\cdot, \cdot) \rightarrow \hat{s}_y(i) = \{\hat{s}_{y_j}\}_{j \in B_i} = g_\phi(s_x, \{m_j\}_{j \in B_i}) \rightarrow \hat{s}_y(1), \dots, \hat{s}_y(M)$

□ **L_2 Loss:** $\frac{1}{M} \sum_{i=1}^M D(\hat{s}_y(i), s_y(i)) = \frac{1}{M} \sum_{i=1}^M \sum_{j \in B_i} \|\hat{s}_{y_j} - s_{y_j}\|_2^2$ (**Latent Space**)

□ **Gradient-based optimization:** θ & ϕ

□ **Exponential Moving Average:** Target Encoder $\bar{\theta}_t = m_t \bar{\theta}_{t-1} + (1 - m_t) \theta_t$ $m_t: 0.996 \rightarrow 1.0$

Implicit Route: Predictive Representation Learning via V-JEPA



□ $T \times H \times W \rightarrow \text{Patch: } 2 \times 16 \times 16 \rightarrow L \text{ Spatio-temporal Tokens} \rightarrow L \times d$

□ Sample y from a video: Sample several (possibly overlapping) spatially continuous blocks with various aspect ratios and repeat the spatial blocks **across the entire temporal dimension of the video**.

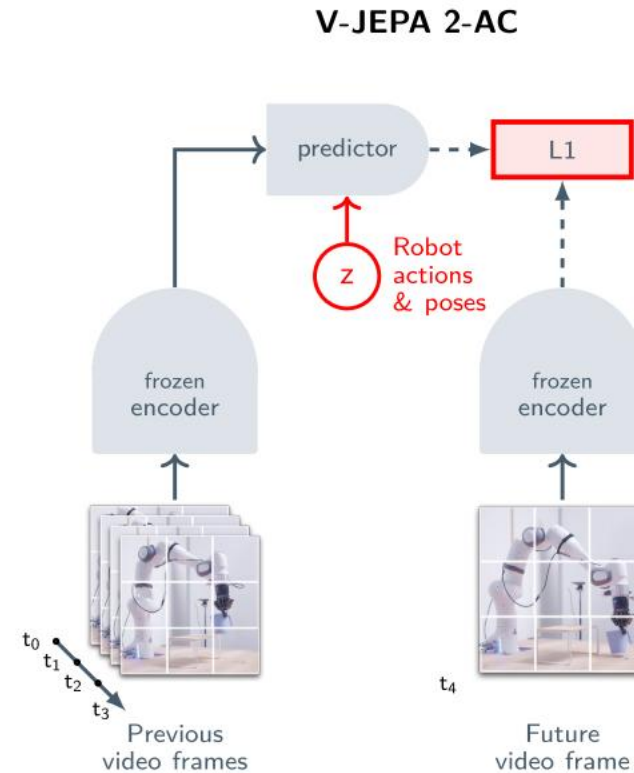
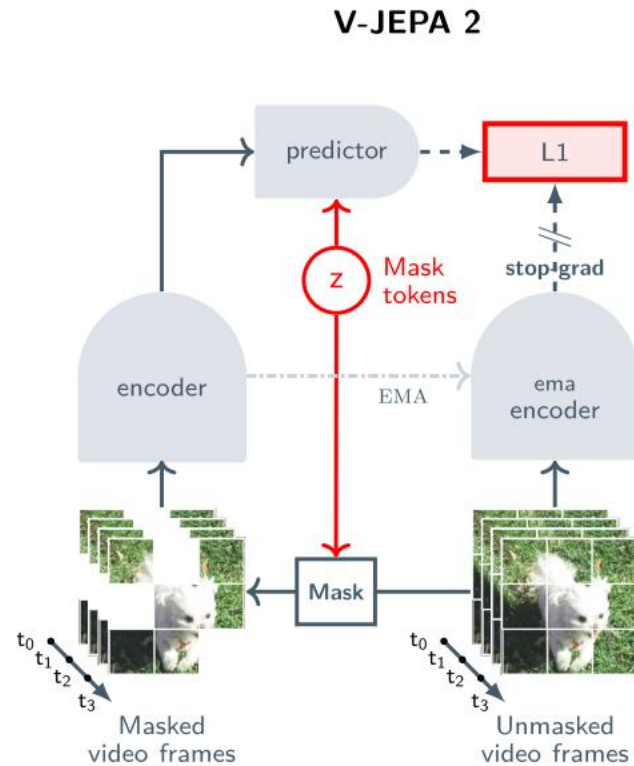
□ **L₁ Loss:** $\min_{\theta, \phi} \|P_\phi(E_\theta(x), \Delta_y) - \text{sg}(\tilde{E}_\theta(y))\|_1$ (**Latent Space**)

V-JEPA 2-AC



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❑ Freeze The Video Encoder

❑ Learn a New Action-conditioned Predictor

❑ Autoregressive Feature Prediction objective: Conditioned on Past Video Frames & Actions & End-effector States → Predict the **Representations of Future Video Frames**

03

Architecture of World Action Models

Two Primary Paradigms: Cascaded WAM vs. Joint WAM



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*These World Action Model architectures can be categorized into two primary paradigms based on their **structural flow** and **corresponding training regimes**.*

Cascaded

Think Before You Act

- 01 **Mathematical Essence:** $\mathcal{L}_{Cascaded} = p(o', a|o, l) = p(a|o', o, l) p(o'|o, l)$
- 02 **Separated Training Process:** The World model and the action decoder are optimized as distinct modules.
- 03 **Representative Models:** **UniPi**, VPP, $\pi_{0.7}$, etc.

Joint

Think While You Act

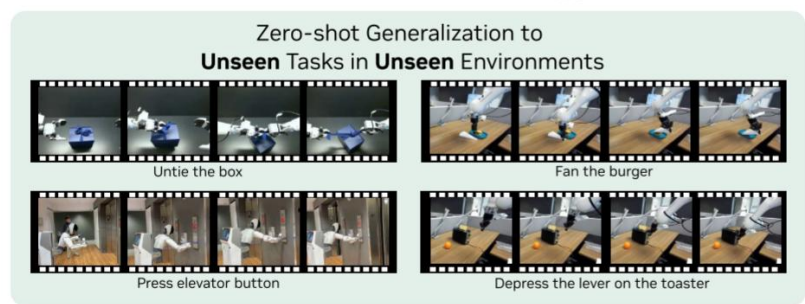
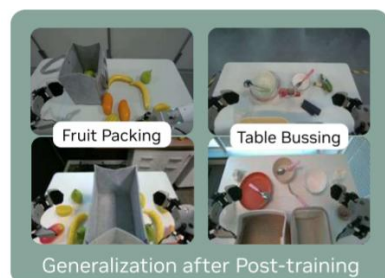
- 01 **Mathematical Essence:** $\mathcal{L}_{Joint} = p(o', a|o, l)$
- 02 **Joint Training Process:** World modeling and action generation undergo joint training under a unified objective, forcing the model to internalize the **causal interdependencies** between **environmental dynamics** and **control signals**.
- 03 **Representative Models:** **DreamZero**, UWM, WorldVLA, etc.

Joint WAM: DreamZero



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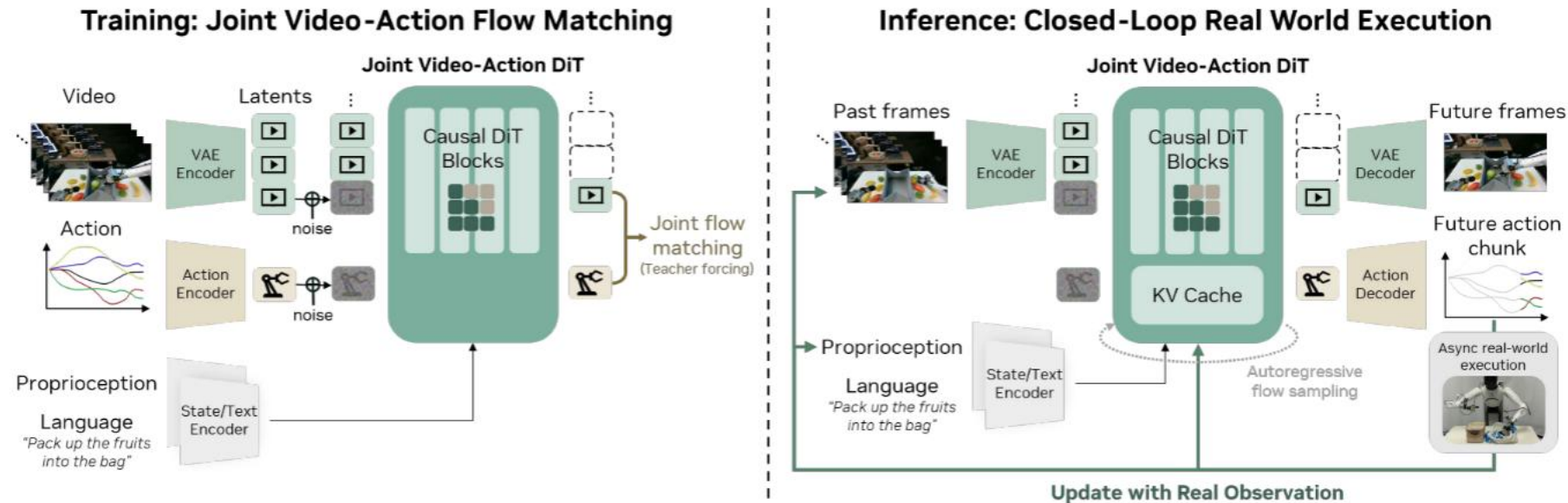
- ❑ Effective learning from diverse, non-repetitive data (**Task Diversity > Repetition**)
- ❑ Open-world generalization (**Zero-shot generalization to new tasks in new environments**)
- ❑ Cross-embodiment learning from video-only data (**humans (12 minutes) or other robots (20 minutes)**)
- ❑ Few-shot adaptation to new robots (**Adapt to an entirely new robot with only 30 minutes of play data**)

Joint WAM: DreamZero



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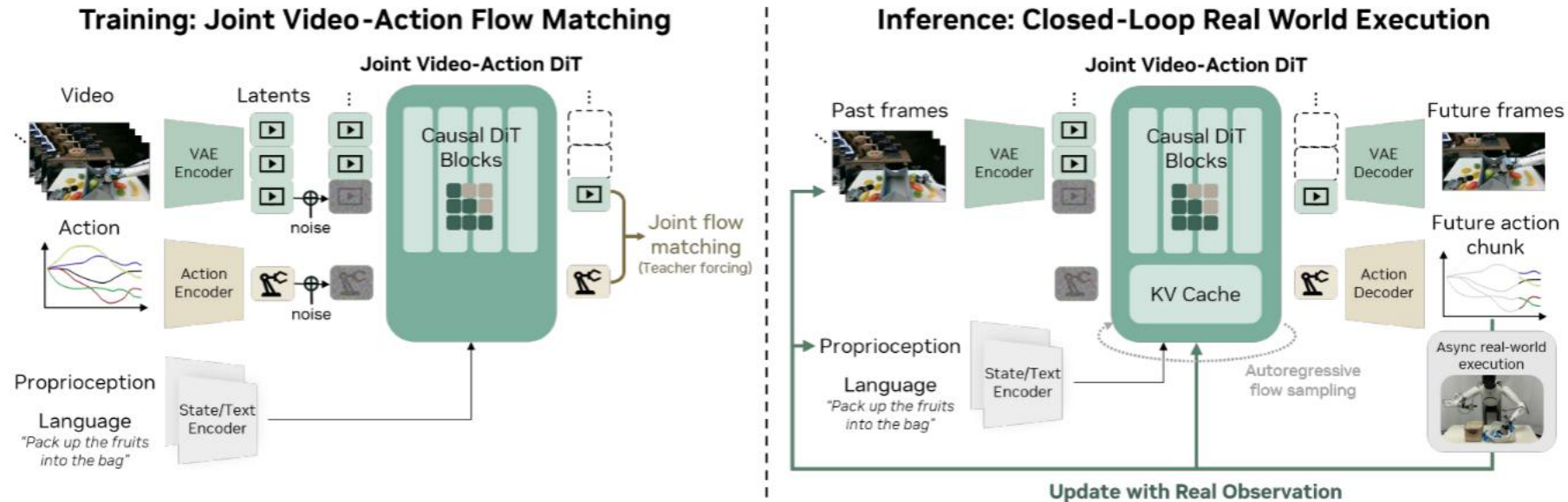
- ❑ Architecture: Autoregressive Diffusion Transformer
- ❑ Multi-view Concatenation: Concatenate all views into a single frame
- ❑ Video → Frozen VAE Encoder → Visual Context
- ❑ Language Instructions → Text Encoder → Text Embedding
- ❑ Proprioception → State Encoder → State Embedding

Joint WAM: DreamZero



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□ Autoregressive Modeling Only for the Video Modality: $\pi_0(o_{l:l+H}, a_{l:l+H} | o_{0:l}, c, q_l) = \underbrace{\pi_0(o_{l:l+H} | o_{0:l}, c, q_l)}_{\text{video prediction}} \underbrace{\pi_0(a_{l:l+H} | o_{0:l+H}, q_l)}_{\text{IDM}} \underbrace{\quad}_{\text{DreamZero}}$

□ Teacher-forcing: Model is trained to denoise the noisy current chunk conditioned on the clean previous chunks $C_k = \{(z_1^j, a_1^j)\}_{j=1}^{k-1}$, where z_1^k and a_1^k are a clean video latent vector and a normalized action.

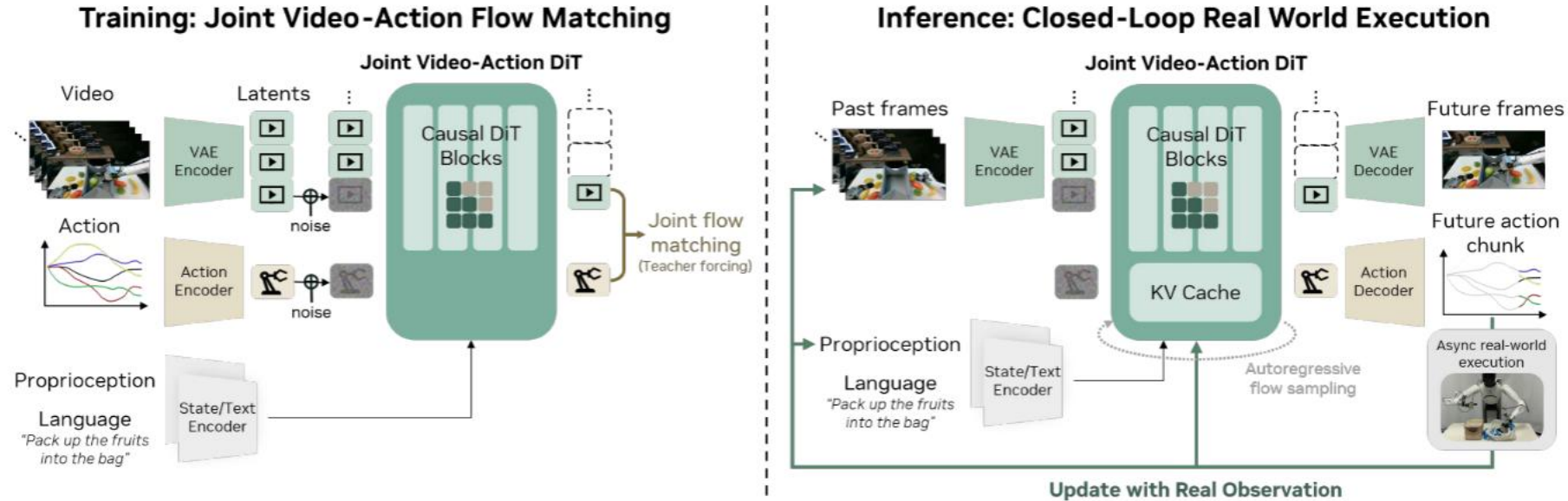
□ Inference: Enables Faster Inference Speed by Utilizing KV-cache

Joint WAM: DreamZero



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□ Linear Interpolations between Clean Vectors and Random Gaussian Noises: $z_{t_k}^k = t_k z_1^k + (1 - t_k) z_0^k$, $a_{t_k}^k = t_k a_1^k + (1 - t_k) a_0^k$, $z_0^k \sim \mathcal{N}(0, I)$, $a_0^k \sim \mathcal{N}(0, I)$

□ Flow Matching Loss:

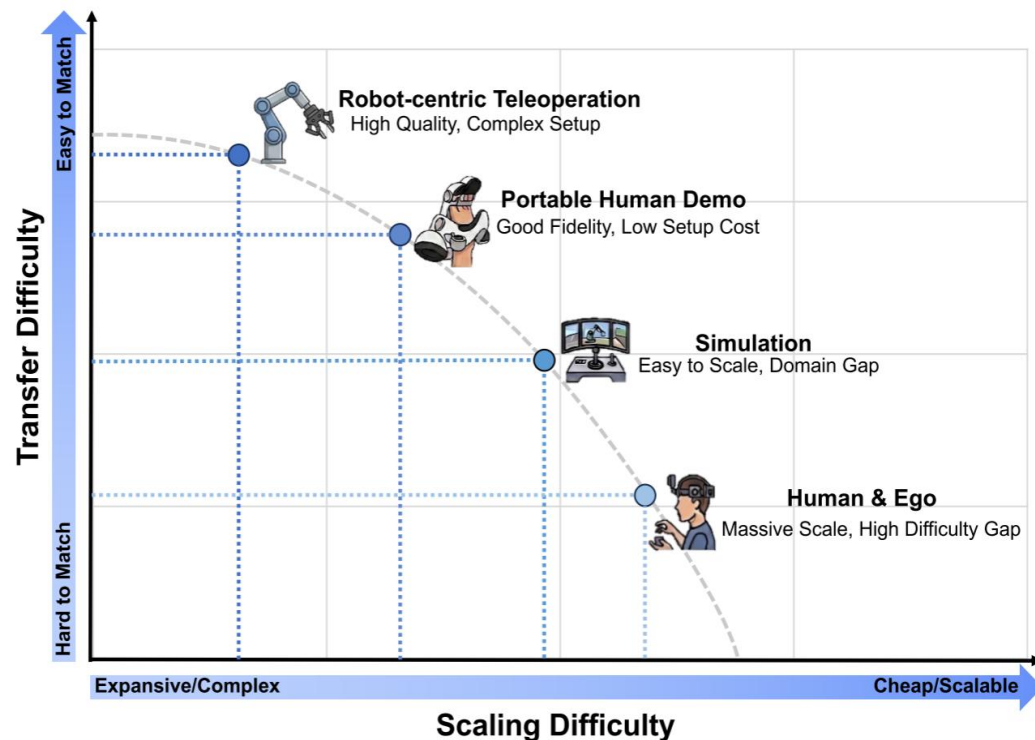
$$\mathcal{L}(\theta) = \mathbb{E}_{z,a,\{t_k\}} \left[\frac{1}{K} \sum_{k=1}^K w(t_k) \|u_{\theta}([z_{t_k}^k, a_{t_k}^k]; \mathcal{C}_k, c, q_k, t_k) - v^k\|^2 \right], \mathcal{C}_k = \left\{ (z_1^j, a_1^j) \right\}_{j=1}^{k-1},$$

$$v^k := [z_1^k, a_1^k] - [z_0^k, a_0^k]$$

04

Data & Evaluation

Embodied Data for training WAMs



❑ High-Fidelity Robot-Centric Teleoperation

- ✓ Core Role: Provides precise kinematic grounding.

❑ Agile Portable Human Demonstrations (UMI-style)

- ✓ Core Role: Acts as a critical bridge connecting human dexterity and real-world interactions.

❑ Highly scalable Simulation Data

- ✓ Core Role: Break the bottlenecks of real-robot data collection.

❑ Broad-Coverage Human- and Ego-Centric Data

- ✓ Core Role: Internalizes unbounded passive world dynamics knowledge via passive visual observations.

Datasets Examples

❑ DROID

DROID Dataset Visualizer (Showing random subset of 1k trajectories from DROID)

Scene Types

- ☒ All (1000)
- ☐ Industrial office (586)
- ☐ Kitchen (252)
- ☐ Office (82)
- ☐ Living room (27)
- ☐ Dining room (17)
- ☐ Bathroom (14)
- ☐ Bedroom (11)
- ☐ Hallway (8)
- ☐ Laundry room (3)
- ☐ Other (0)

Object Types

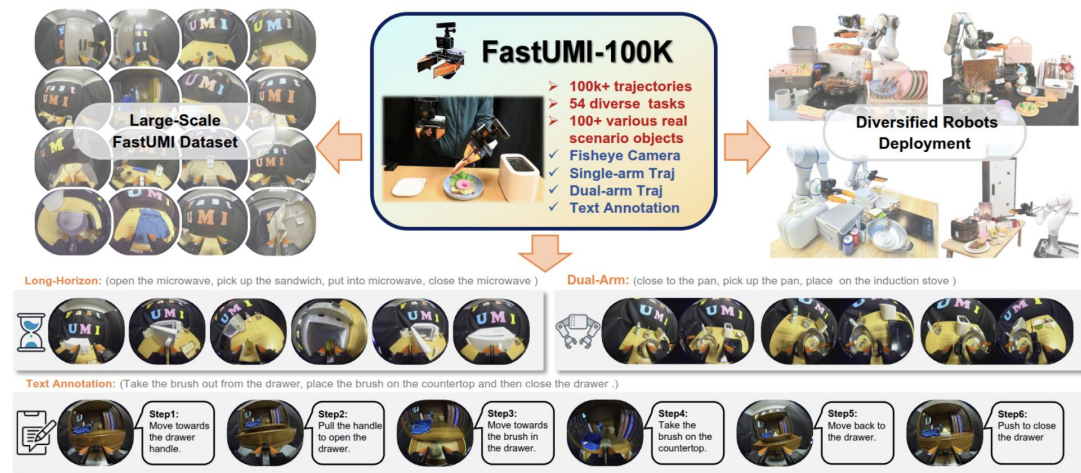
- ☒ All (1000)
- ☐ Marker (163)
- ☐ Cloth (72)
- ☐ Cup (36)
- ☐ Object (32)
- ☐ Bottle (31)
- ☐ Block (30)
- ☐ Lid (24)
- ☐ Drawer (23)
- ☐ Bowl (20)
- ☐ Other (569)

Task Types

- ☒ All (1000)
- ☐ Put (237)
- ☐ Move (141)
- ☐ Pick (133)
- ☐ Remove (95)
- ☐ Take (57)
- ☐ Close (55)
- ☐ Open (54)
- ☐ Place (28)
- ☐ Turn (26)
- ☐ Other (174)

Showing 40 matches randomly sampled from 1000 total matches

❑ FastUMI-100K



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❑ RoboTwin 2.0

50 Dual-arm Tasks

731 Diverse Objects

5 Embodiments

Rich Diversity

Benchmark

Large Dataset Scale

RoboTwin

Performance

Sim2Real

4.7x ↑ 30% 42%

9% 30% 42%

Only Real-World Zero-shot Few-shot

Auto Code Generation

71.3% (ours)

60.4% 63.9%

47.4%

RoboTwin 1.0 Vanilla
RoboTwin 1.0+FB
RoboTwin 1.0+FB+MLLM
RoboTwin 2.0

Domain Randomization

Randomization

Unseen Real-World

Open Source

RoboTwin Object Dataset

Community LeaderBoard

Deploy Diverse Policies

Pre-Collected 100k+ Traj.

Friendly Codes & Tutorial

Generate Any Dual-Arm Robot Manip Data

❑ Ego4D



How to Validate Real-World Effectiveness of WAMs?



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❑ World Modeling Capability

❑ Visual Fidelity:

- Low-Level Fidelity: PSNR, SSIM
- Perceptual and Semantic Consistency: LPIPS, DreamSim, DINO Similarity
- Distribution-Level Realism and Temporal Quality: FVD

❑ Physical Commonsense: WorldModelBench, WorldScore, etc.

❑ Action Plausibility: WorldSimBench, etc.

❑ Action Policy Capability

❑ Task Success Rate

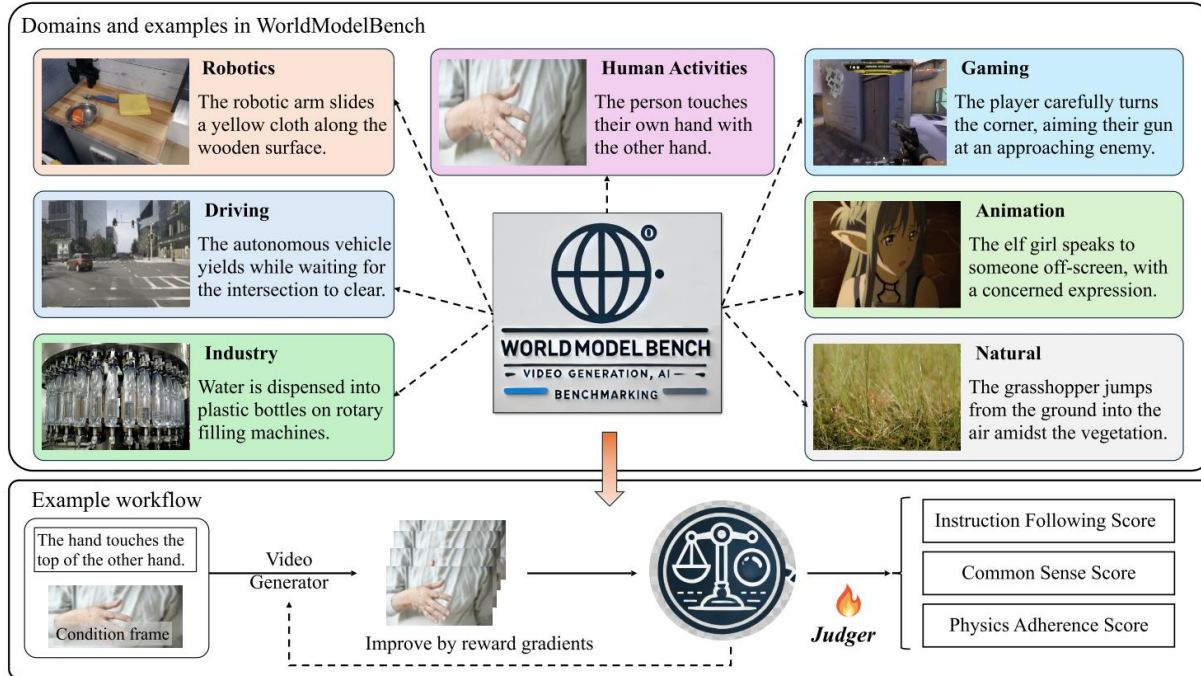
❑ Average Task Progress

Physical Commonsense: WorldModelBench



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A robotic arm hovers above a wooden table, ready to interact with the towel.



Model A



Model B



VBench

	Smoothness	Consistency	Image Quality	Win Rate
Model A	✓	✓	✓	0.5
Model B	✓	✓	✓	0.5



Ours

	...	Newton's Law	Gravity	Win Rate
Model A	...	✓	✗	0.2
Model B	...	✓	✓	0.8

❑ **Details:** 350 image-text condition pairs, 7 application-driven domains, 56 diverse subdomains, supports T2V & I2V, 67K human labels

❑ **Assessment Principles:** Instruction Following (Scores 0-3) & Physics Adherence (Total Scores 0-5) & Commonsense (Frame-Wise Quality & Temporal Quality) (total scores 0-2)

❑ **A Reliable World Modeling Judger (Reward Model)**

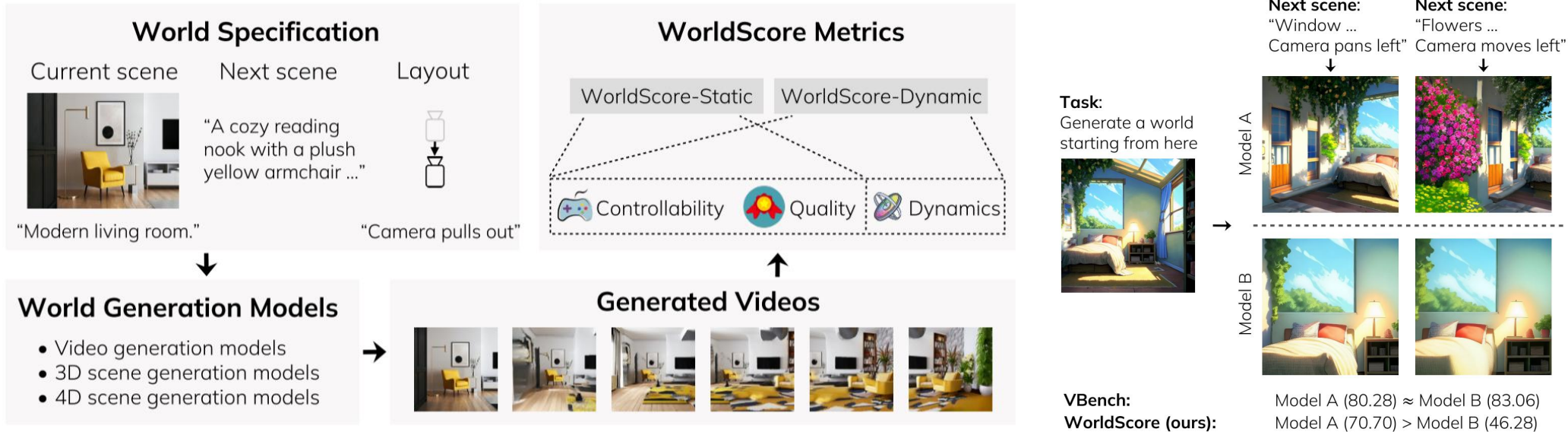
$$J(\theta) = \mathbb{E}_{c \sim D_c, x_0 \sim p_\theta(x_0|c)} \left[\sum_{g \sim G} R(x_0, c, g) \right]$$

Physical Commonsense: WorldScore



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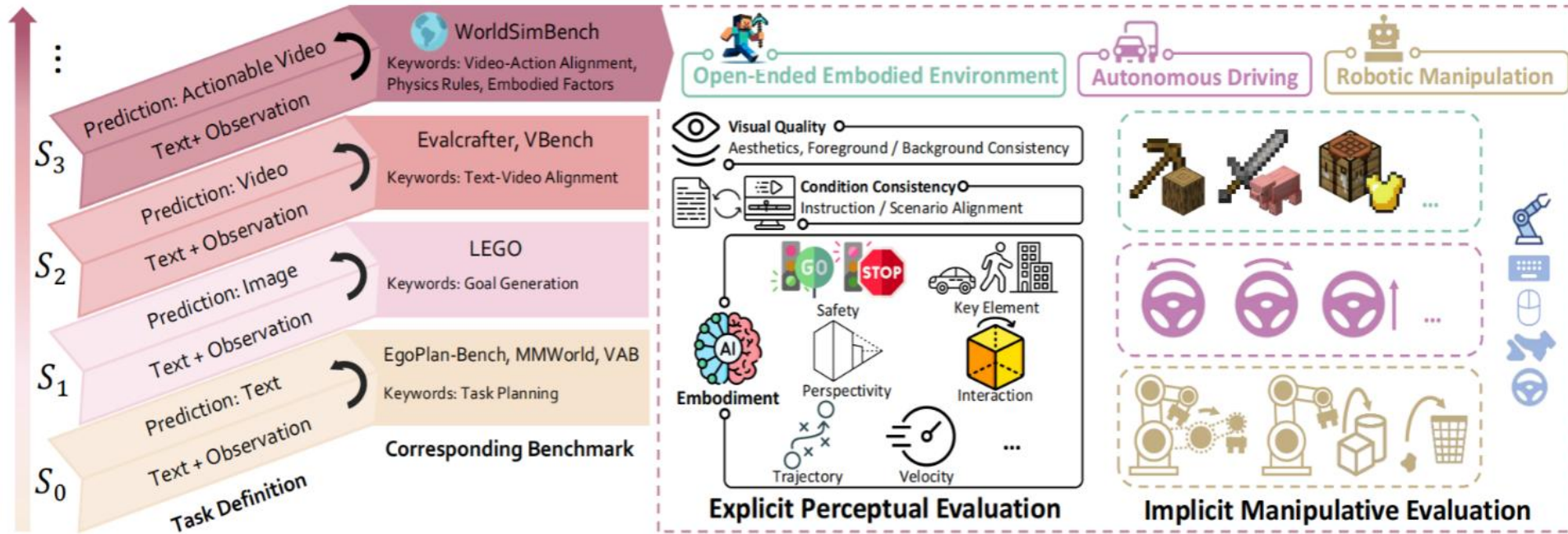
- ❑ **Motivation:** Existing benchmarks lack explicit spatial-layout control, camera specifications and reference images, which limits evaluations to single scenes.
- ❑ **World Generation** → A Sequence of Next-Scene Generation Tasks. (current scene, next scene, layout)
- ❑ **Evaluation Scope:** 3D, 4D, T2V, I2V
- ❑ **10 Metrics:** **Controllability** (Camera Controllability, Object Controllability, Content Alignment), **Quality** (3D Consistency, Photometric Consistency, Style consistency, Subjective quality), **Dynamics** (Motion accuracy, Motion magnitude, Motion smoothness)
- ❑ **By Application Scenarios:** WorldScore-Static & WorldScore-Dynamic

Action Plausibility: WorldSimBench



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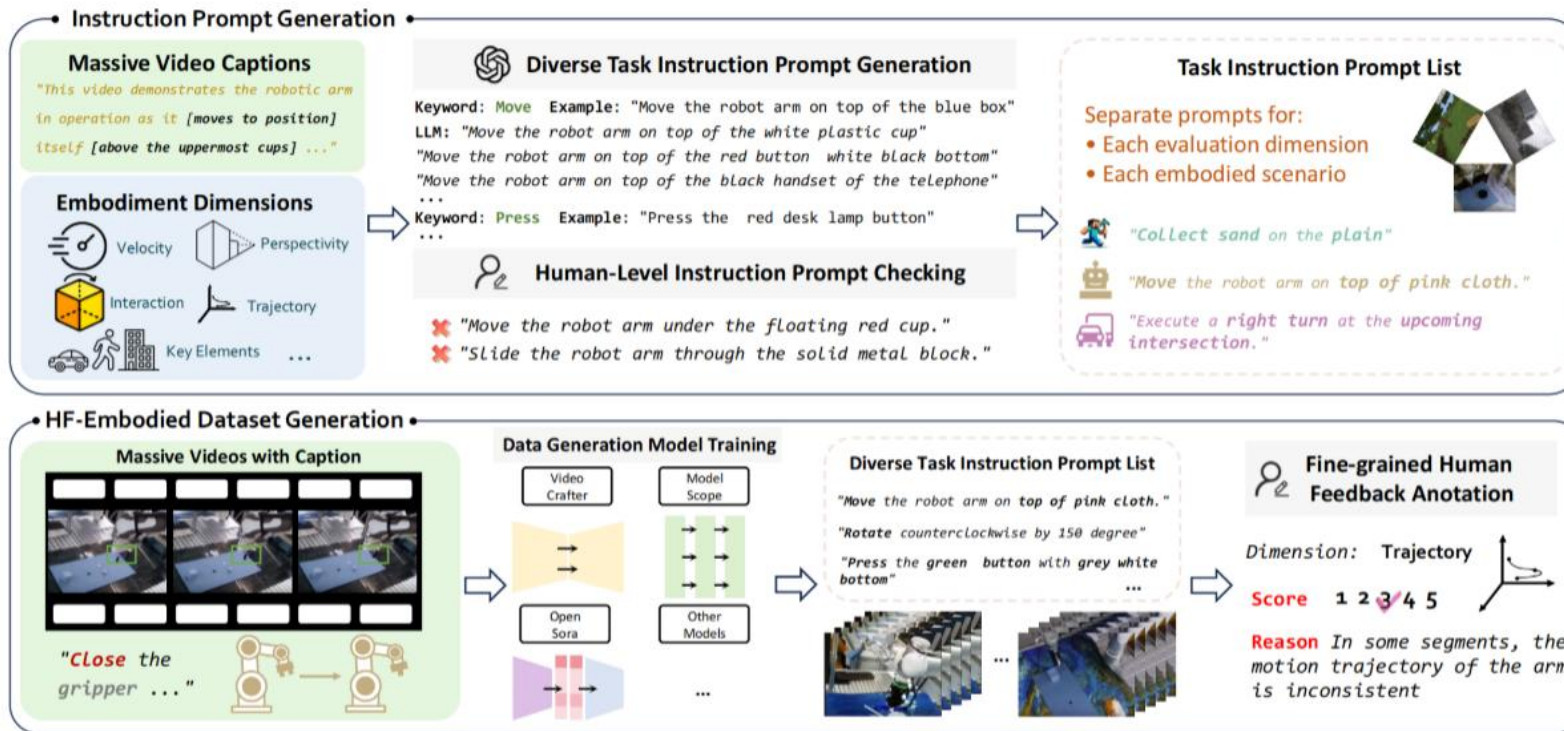
- ❑ Hierarchical Capabilities of the Predictive Models: $S_0 \rightarrow S_3$
- ❑ Dual Evaluation Framework: Explicit Perceptual & Implicit Manipulative
- ❑ Three Embodied Scenarios: Open-Ended Embodied Environment, Autonomous Driving, Robot Manipulation
- ❑ Focus: Video Generation & Action Transformation

Action Plausibility: WorldSimBench



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✓ **Explicit Perceptual Evaluation**

❑ **Instruction Prompt Generation**

❑ **HF-Embodied Dataset Generation**

❑ **Video Assessment Dataset: HF-Embodied Dataset (35,701 entries)**

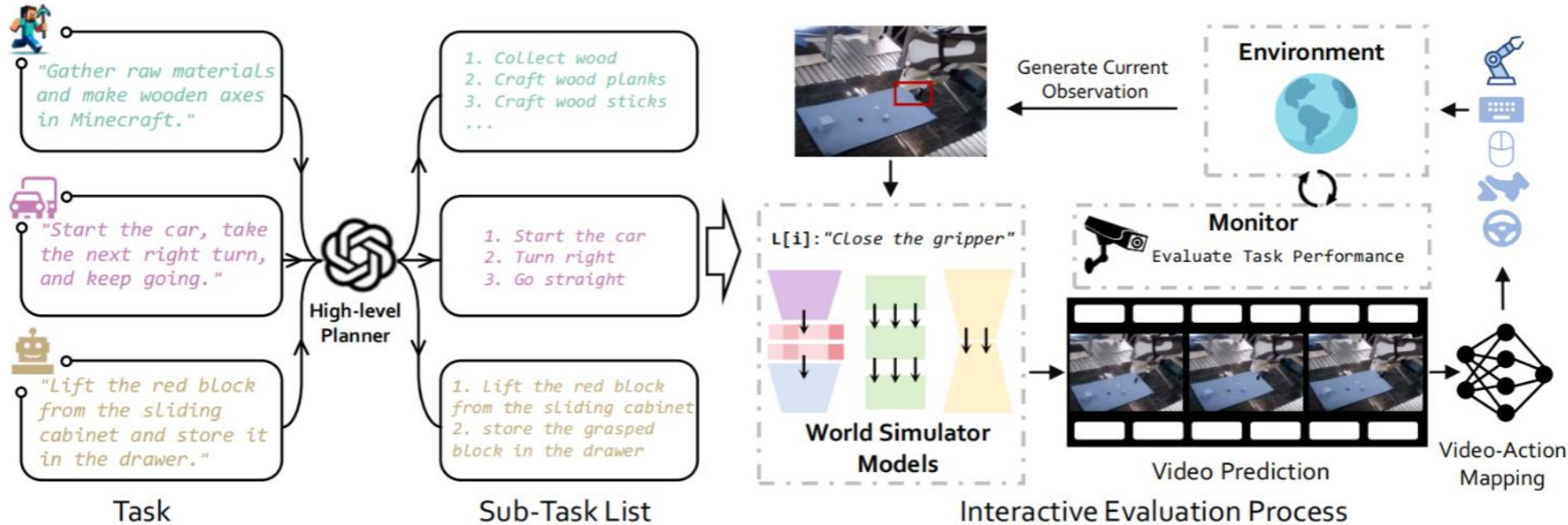
❑ **Train a Human Preference Evaluator:** Aligns with human perception and explicitly assesses the visual fidelity of World Simulators.

Action Plausibility: WorldSimBench



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✓ **Implicit Manipulative Evaluation**

- ❑ Assess the Video-Action Consistency of World Simulators
- ❑ Embodied Tasks in Different Scenarios → Executable Sub-tasks
- ❑ Current Instructions + Observations → Predicted Videos
- ❑ Video → Pre-trained IDM or a Goal-based Policy → Action
- ❑ Evaluation: Success Rates of Various Embodied Tasks



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THANKS